



FAPS

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Circular economy for electric machines

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

Electric machines contain a significant amount of valuable materials making circular economy relevant for economic, ecological and geopolitical reasons.

Circular economy includes reuse and recycling, with the metallic components of electrical machines being the main value carriers.

After attending the lecture unit "Circular economy for electric machines", you should know:

- **the definition of circular economy and the differentiation between “reuse” and “recycling”**
- **the advantages and challenges in establishing a circular economy for electric machines**
- **the main value carriers of electric machines**
- **the process chains for the recovery of relevant materials and components and the related methods**
- **and the strategies for implementing design for recycling.**

Graphic: Stokkermill



Graphic: Jansen Recycling



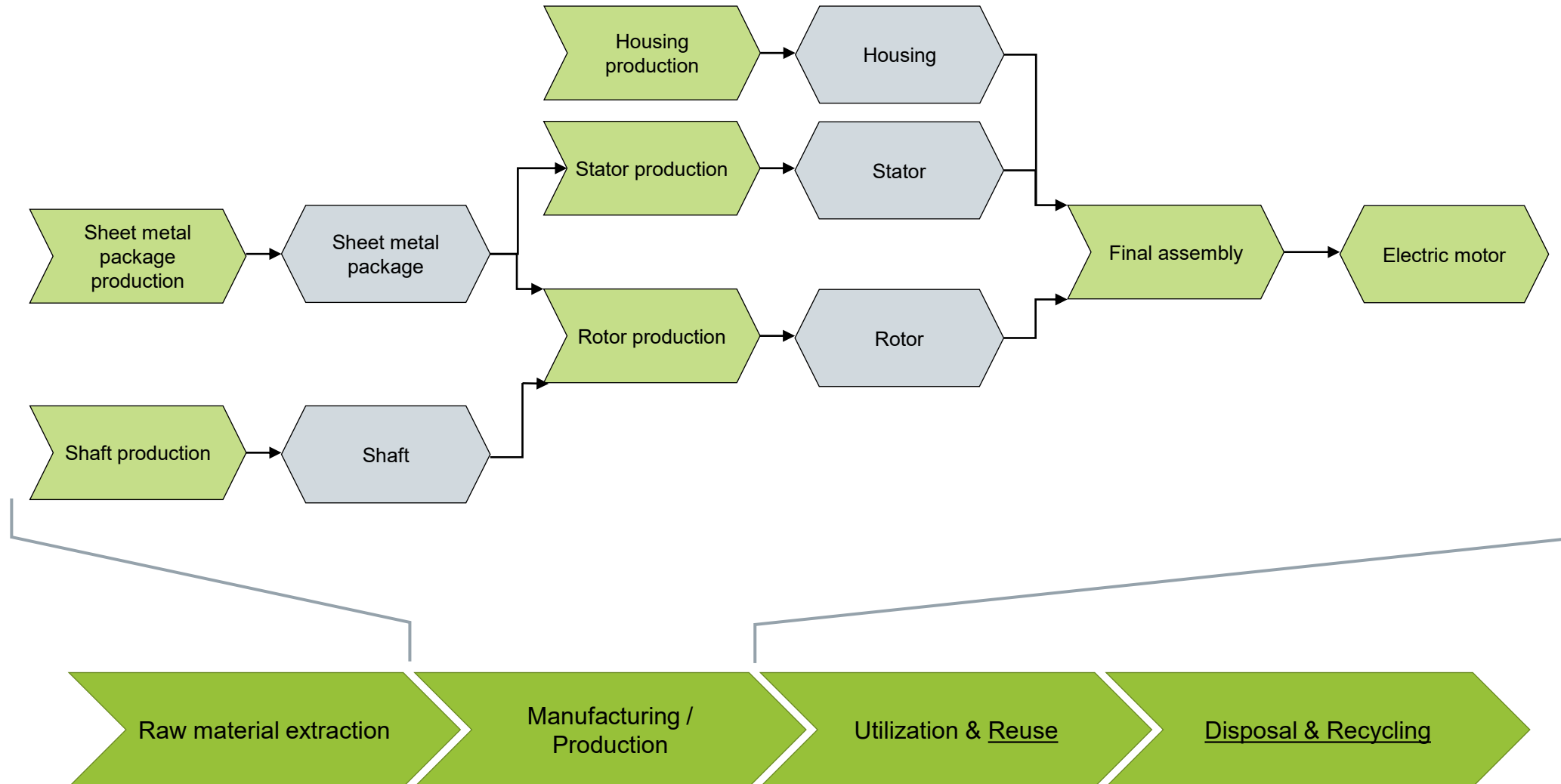
Graphic: Recycling magazine



Graphic: Umweltbundesamt



The process chain for manufacturing an electric motor comprises a large number of process steps and material elements that are closely linked to the circular economy.



This lecture unit provides an overview of the motivation and technologies for processing of end-of-life electric machines.

- **General overview of circular economy and definitions**
- Recycling of electric machines
- Reuse, repair and refurbishment of electric machines
- Design for Recycling

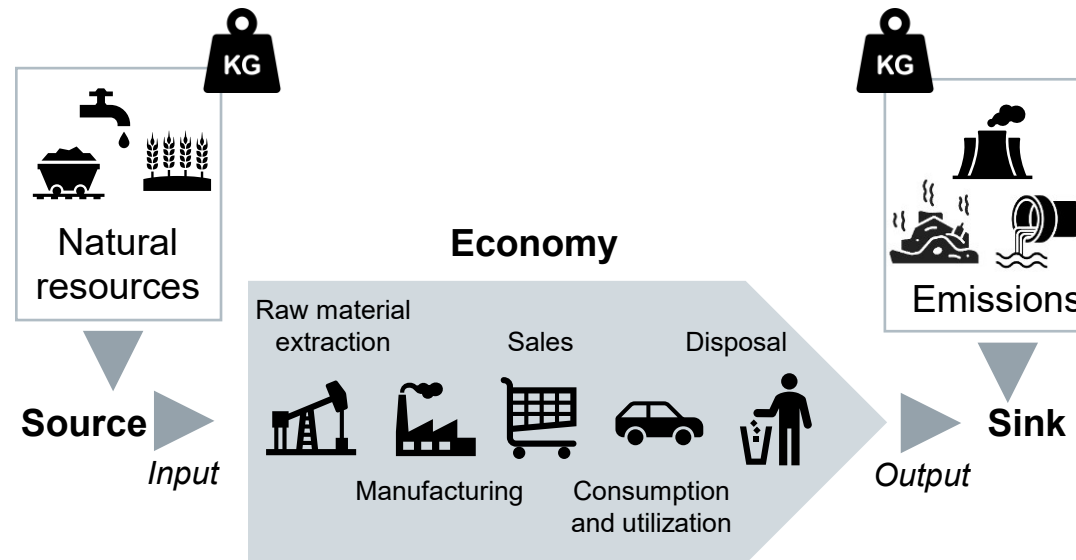


Graphic: Jansen Recycling

The goal of the circular economy is to minimize resource consumption, waste production, emissions, and energy waste.

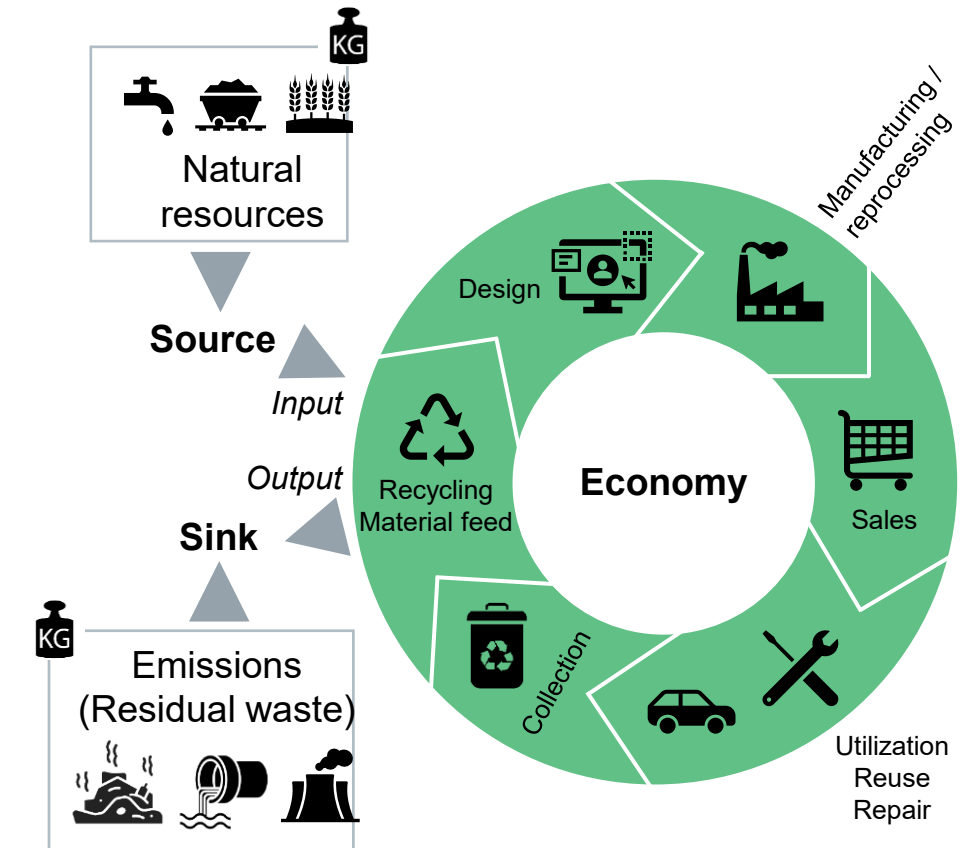
Linear economy

After a single use, materials end up in waste streams (“open loop” - landfill / incineration)

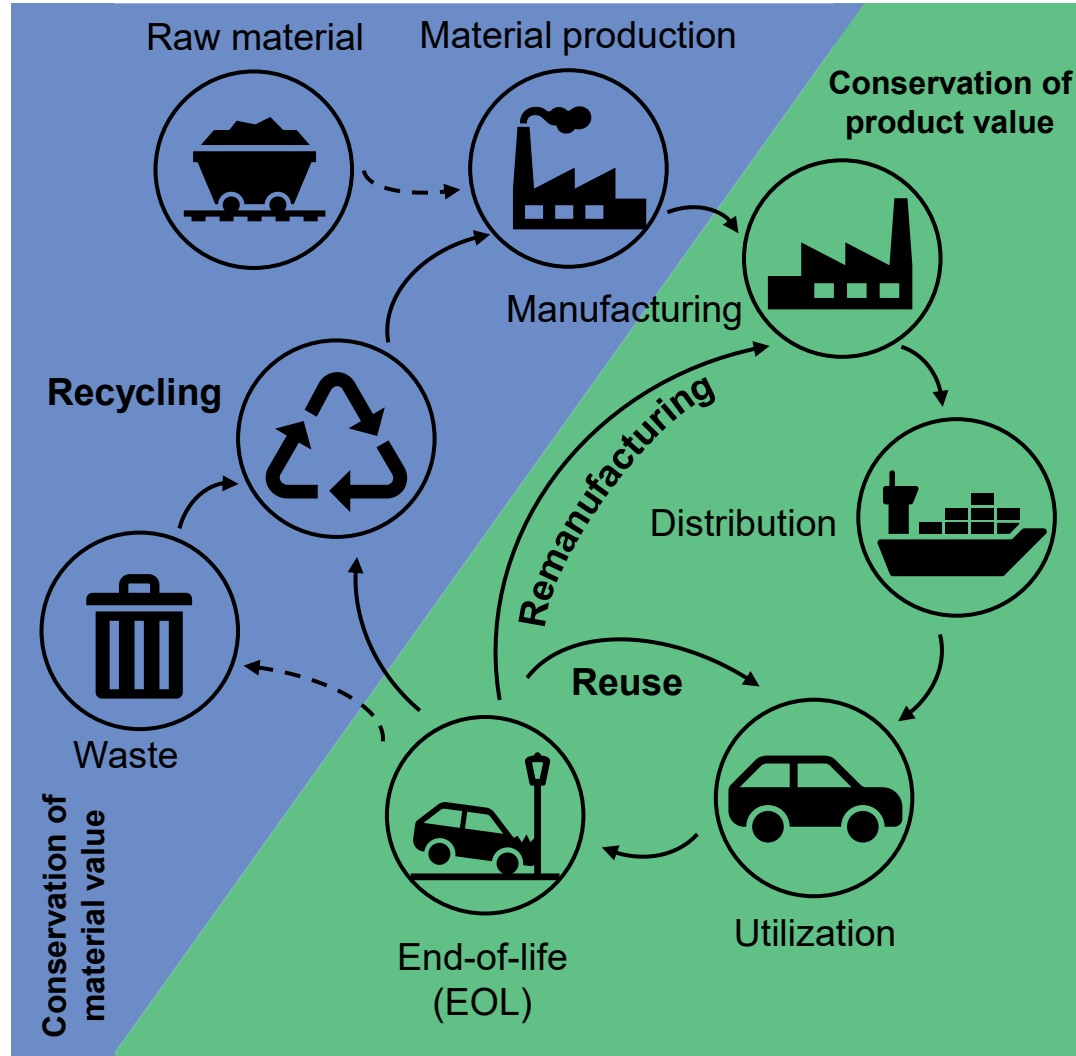


Circular economy

Material and product cycles are largely closed (“closed loop”)



Circular economy comprises recycling and reuse, which differ in the way an end-of-life product is used.



Definitions

■ Circular economy:

Avoidance and utilization of waste

■ Recycling:

Any recovery operation by which waste is converted into products, materials or substances, either for the original purpose or for other purposes.

■ Reuse & Remanufacturing:

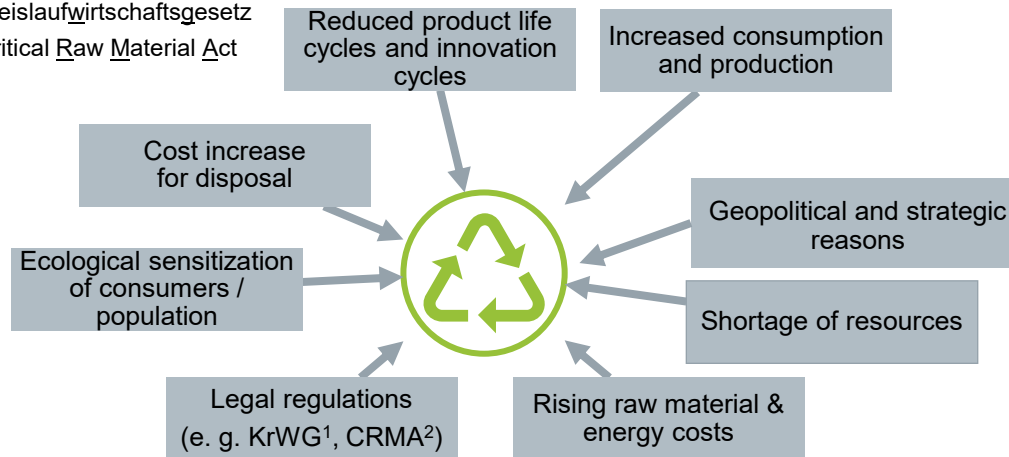
Any process by which products or components that are not waste are reused for the same purpose for which they were originally intended. In this process, a used device is brought up to the quality standard of a new device.

The importance of circular economy is continuously growing due to various factors and is key to a sustainable economic system.

Motivation for circular economy

¹ Kreislaufwirtschaftsgesetz

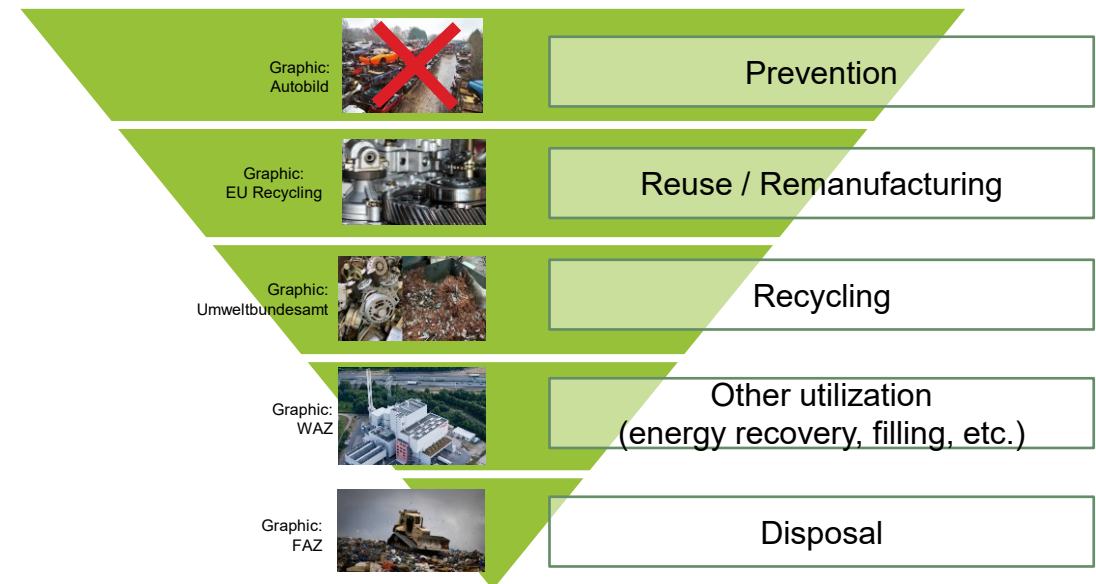
² Critical Raw Material Act



Stakeholders have individual motives for establishing circular economy.

Motives	Relevant stakeholder
Direct economic benefit: Revenue through residual value and avoidance of disposal costs	Individuals & companies
Conservation of resources (Long-term economic & socio-political impacts)	Especially large companies (securing production resources)
Reduction of pollutant emissions (Long-term economic & socio-political impacts)	Whole society

Waste hierarchy



Selection of the method for handling waste based on:

- the expected emissions
- the degree of conservation of natural resources
- the energy to be used or recovered
- the accumulation of pollutants in products and waste

There are several Re-X strategies that allow for the recovery of products, components or materials and must be selected on a case-by-case basis.

Re-X strategies for electric machines

■ Re-use

- Use of motor for same or other application (sometimes less challenging)
- Motor is often not the reason for failure / end of use of EOL product

■ Repair

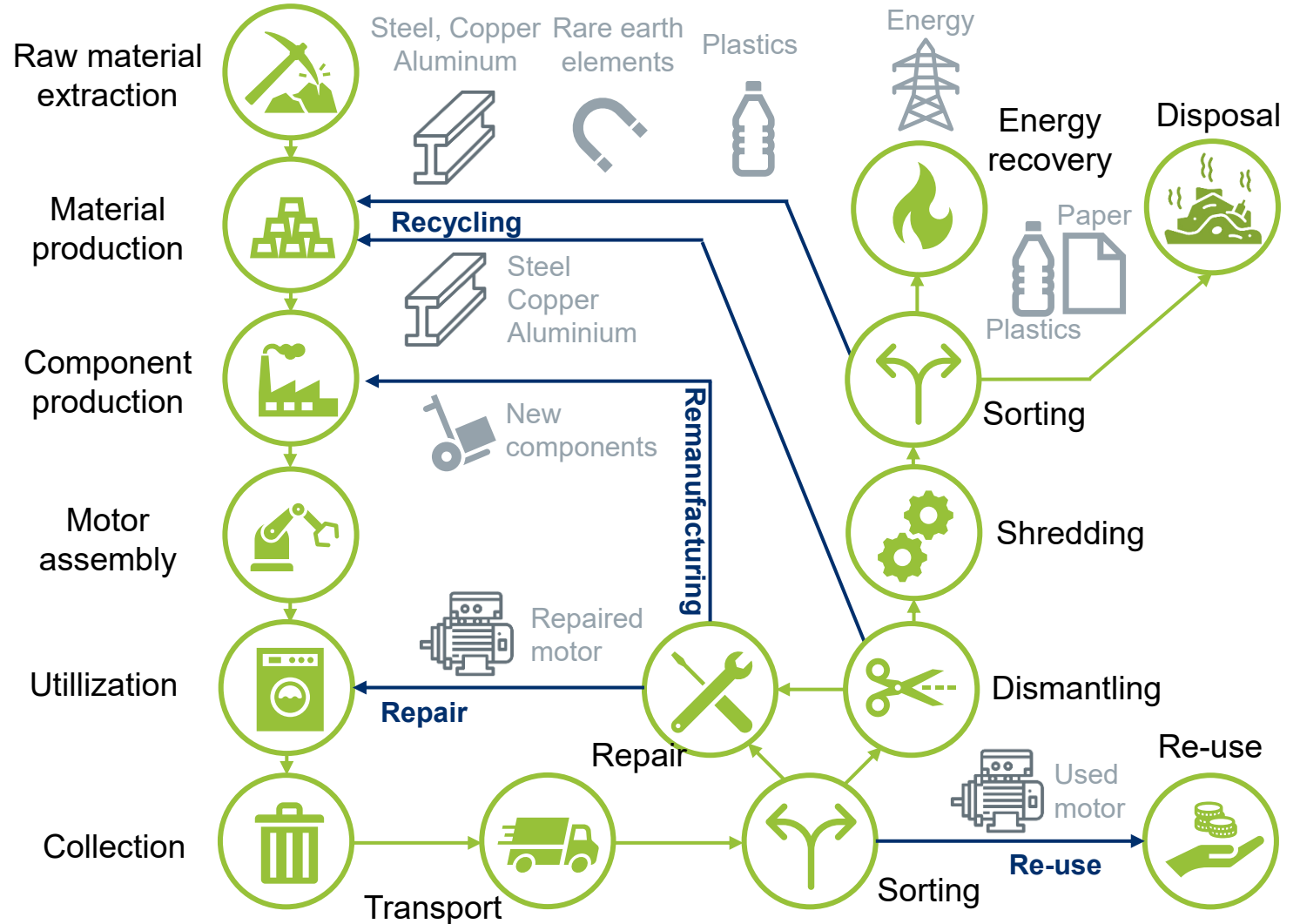
- Cleaning and replacement of worn components (e.g. bearings)
- Life of repaired engine not fully restored

■ Remanufacturing

- Reconditioning and usage of all relevant parts (esp. active parts rotor and stator)
- Challenges: varying degrees of wear, difficulty of non-destructive disassembly and variance in motor topologies

■ Recycling

- Recovery of relevant materials (esp. copper, aluminium, steel)
- High recovery rate and purity of output materials is essential



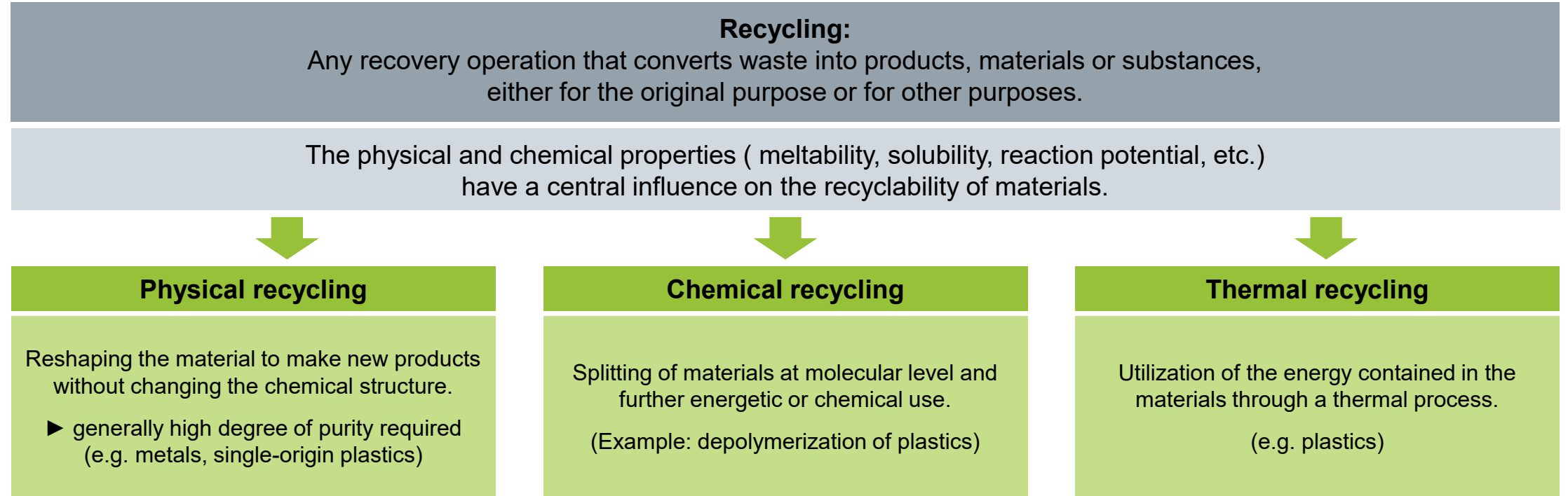
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- General overview of circular economy and definitions
- **Recycling of electric machines**
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- Design for Recycling



Graphic: Jansen Recycling

Recycling allows secondary raw materials to be extracted from waste through treatment processes.



A differentiation can also be made based on the quality of the recycled product:



The recycling of EOL electric machines is driven by economic, ecological and social factors.

Economic factors



- High scrap value
 - End-of-life (EOL) electric machines contain valuable materials
 - For metals, usually no loss of quality of recycled materials
 - Increasing demand for relevant materials (e. g. due to e-mobility and wind power)
- Conservation of limited resources
- Reduced supply chain risk due to reduced dependency from imports

Ecological factors



Reduced environmental & climate footprint

- Mining often has poor environmental footprint (land use, energy and resource consumption, pollutants, impact on ecosys.)
- Energy saving compared to primary production of raw materials
- Reduced greenhouse gas emissions
- Reduction of electronic waste
- Neutralization of harmful substances

Social factors

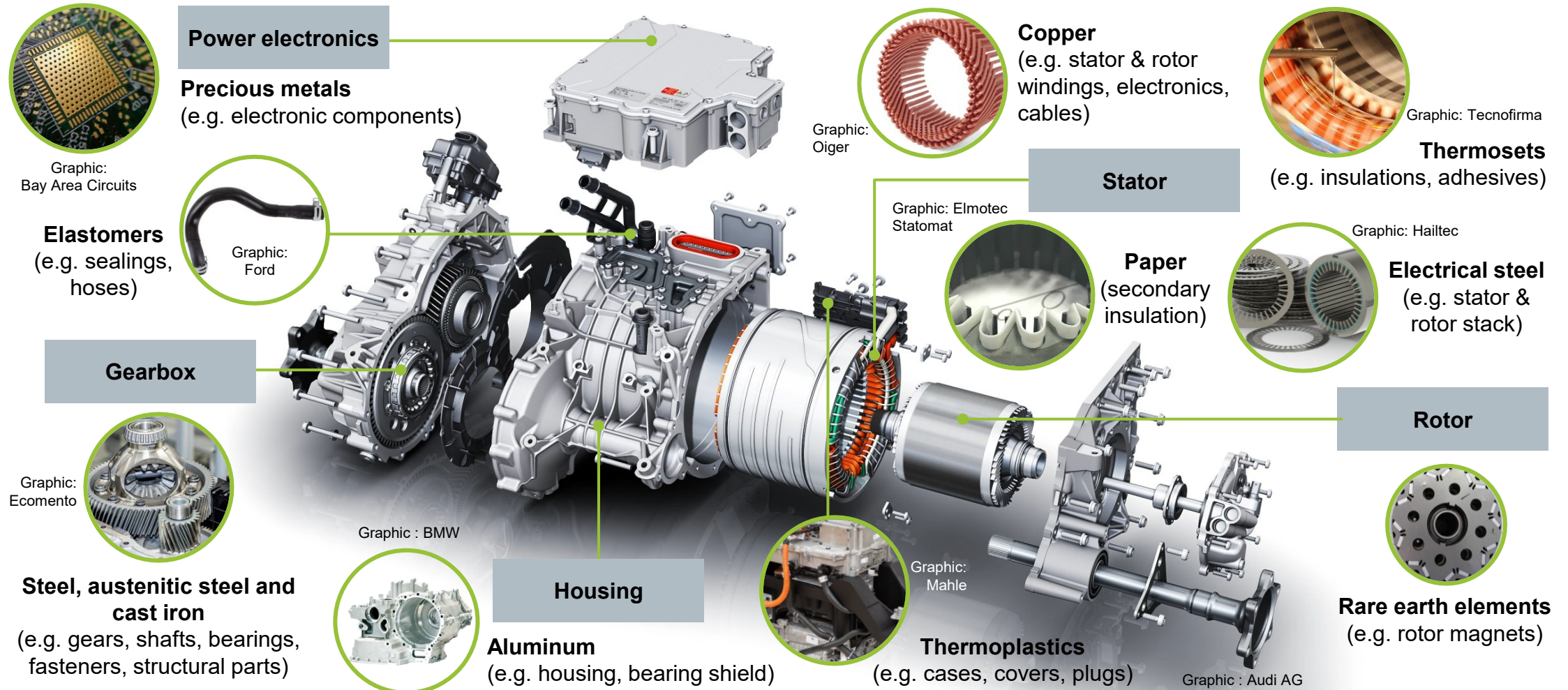


- Social problems caused by the primary production of raw materials and by the export and improper processing of e-waste.
 - Health impacts due to pollution and lack of occupational safety
 - Precarious employment and exploitation
 - Political instability, conflicts, human rights violations and corruption
- Creation of new jobs by recycling industry

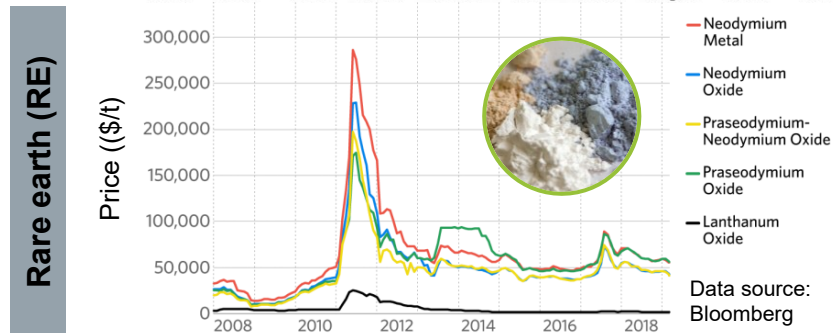
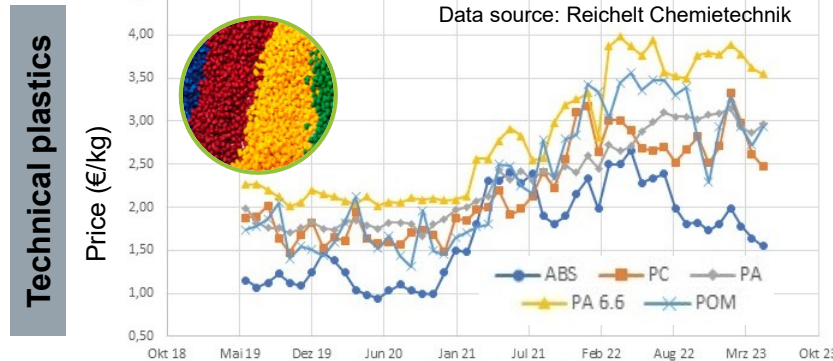
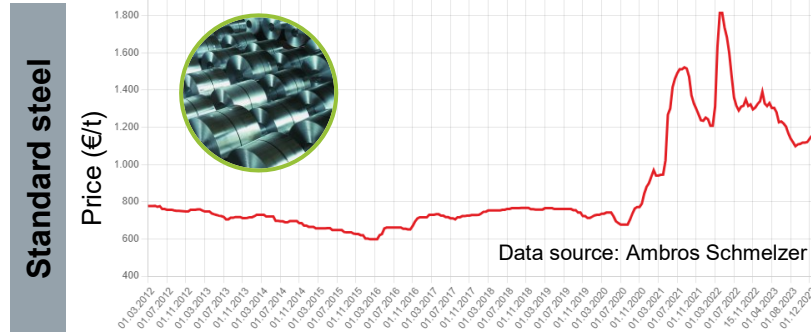
Compliance with legal regulations: Many countries have introduced laws and regulations for the recycling of electric and electronic waste

Electric drives are primarily made of metals, which have a high material value and are well suited for recycling.

Example: automotive traction drive



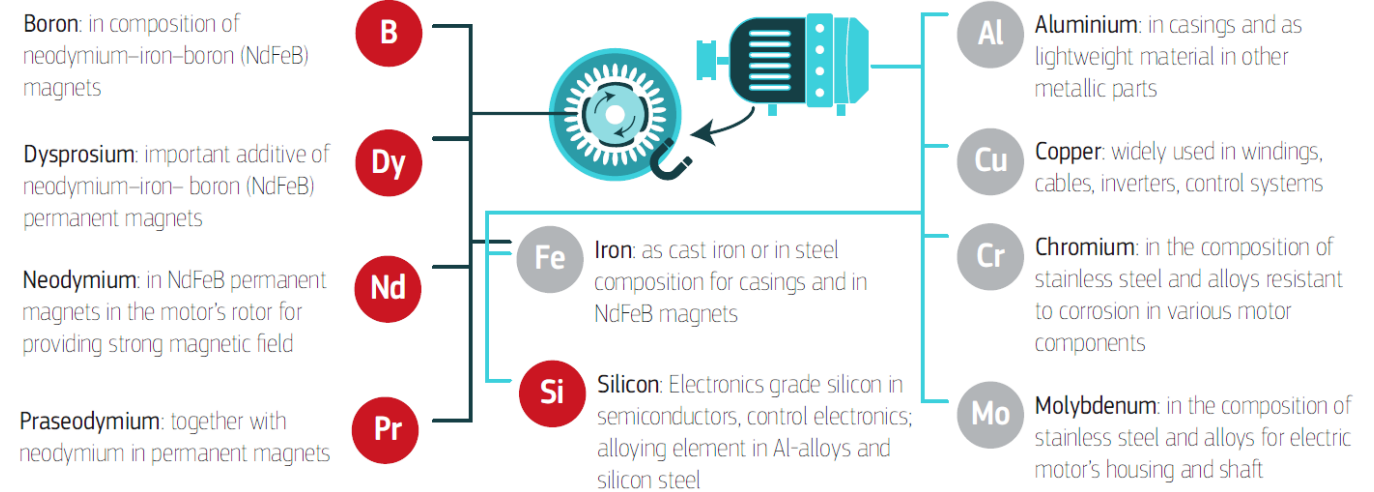
The price trend for many relevant raw materials is a major driver for recycling efforts.



- The occurrence and content of valuable raw materials in electric machines varies depending on the type (e.g. RE magnets only in PM motors)
- Copper from the windings is often the main source of value
- The material value of end-of-life products is subject to strong fluctuations
- Some relevant raw materials are traded on the stock exchange on a daily basis, which affects recycling (e.g. Cu, Al, Sn, Ni, Ag, Au)

Some of the materials used for electric machines are considered as critical raw materials due to their relevance for key technologies and their supply chain risk.

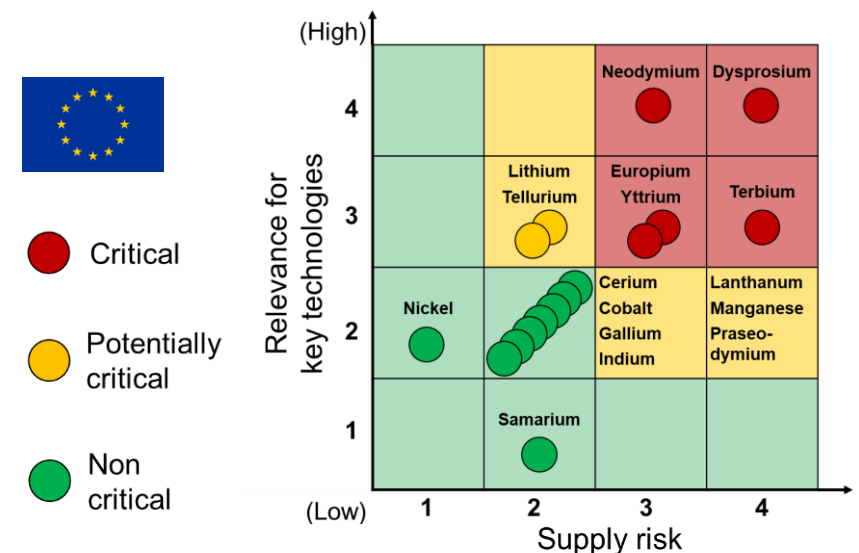
- Critical raw materials used in electric machines include:
 - Rare earth elements (Dy, Nd, Pr, Tb, Ho, Sm) and Boron (B) used for magnets
 - Silicon (Si) used for electronic components
- Copper (Cu) and some alloy elements for steel (e.g. Mo, Cr) are strategic important materials with rising prices
- Many electric devices include also batteries with further critical raw materials (e. g. C, Co, Li, Mn, Nb, Ni, Ti)



Graphic: (Bobba, et. al. 2020)

Critical Raw Materials Act (CRMA):

- Increased sensitivity because of political tensions and pandemics
- European Union identified and ranked most critical raw materials
- Definition of specific measures planned:
 - Development of European extraction and processing capacities
 - Supplier diversification
 - Recycling within the EU will become mandatory (Aim for 2030: 15% - 30% of annual consumption)



The recovery of many materials from e-waste works reliably, but there are major challenges when deviations from the standard process occur.

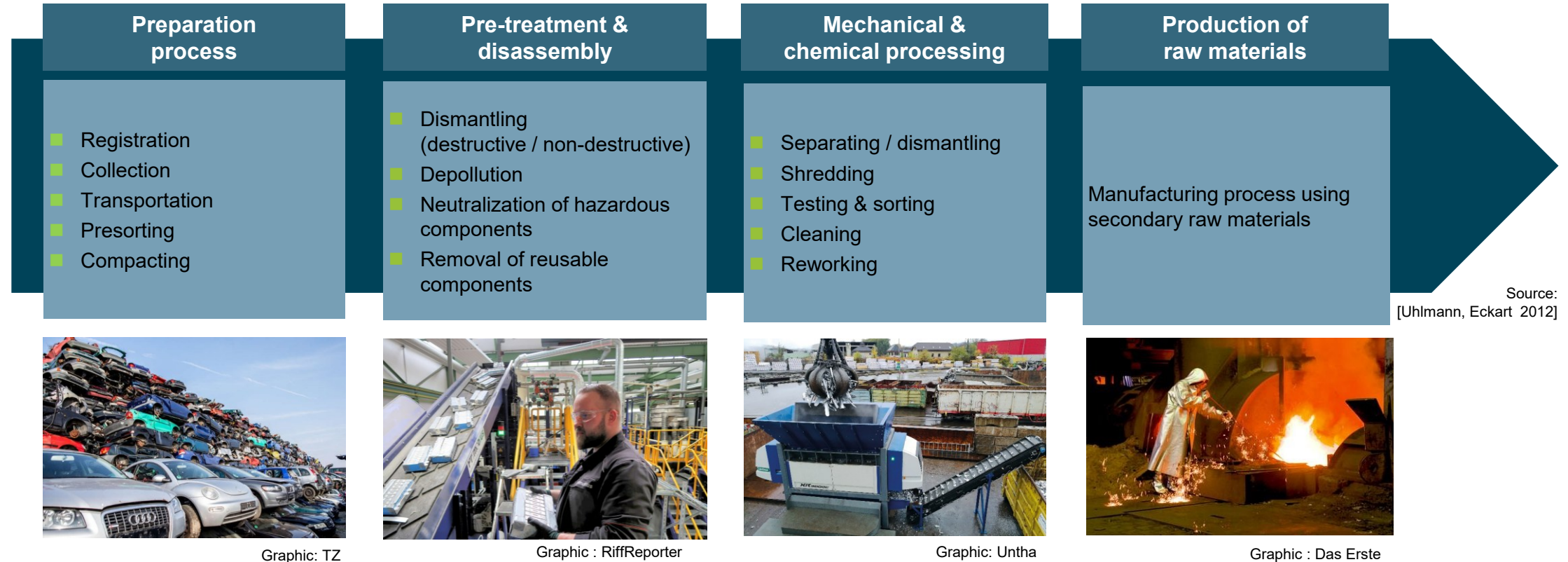
Challenges for the recycling of EOL electric machines

- Very high product variance
- Poor dismantlability
- Poor accessibility of electric machines in EOL products
- Bad product condition (e. g. damages, dirt)
- Lack of information for EOL products
- Automation of dismantling operations is difficult to realize
- Complex pre-treatment if pollutants and harmful components are integrated in EOL products
- Lack of technologies for some recovery steps
- High throughput required for economic efficiency
- Lack of business cases
- Fluctuations in material prices
- High effort for logistics
- Legal regulation



Recycling is a multi-stage process that requires organizational and technical prerequisites.

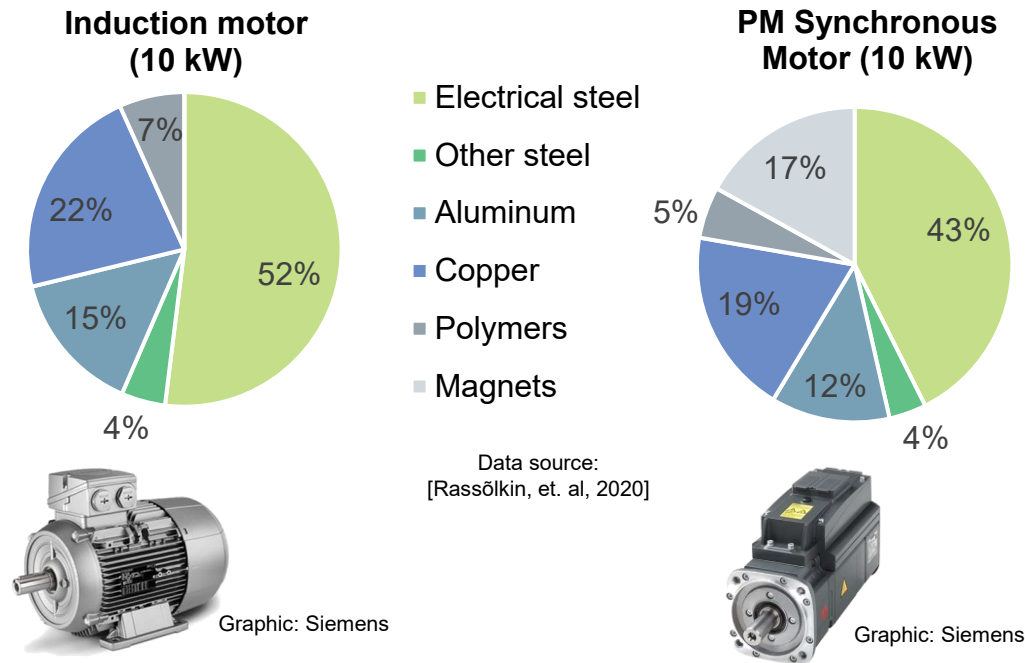
General process chain for recycling



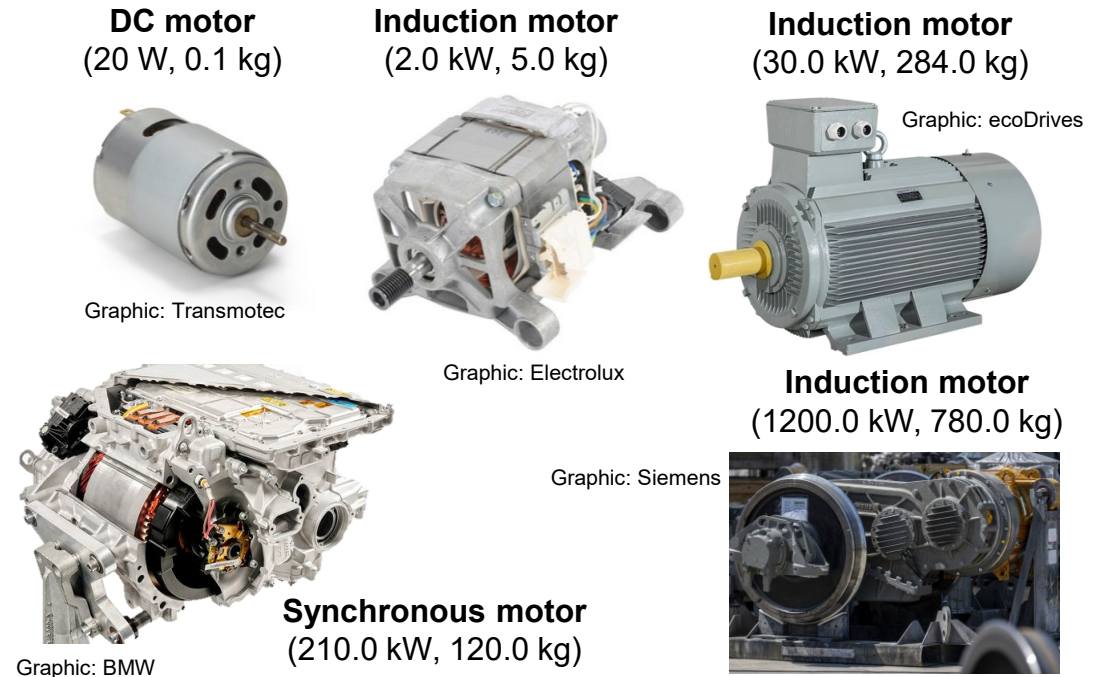
- The recycling process covers the sequence from the generation of waste to the manufacturing of a new product
- The four stages are run through partially or completely

Electrical machines have a very high variance in terms of motor type, size, product volumes and material proportions which affects the recycling approach.

Motor type



Motor size



- Recycling gets more attractive with a higher content of valuable materials such as copper and RE magnets (e. g. rare earth magnets can account for up to 50% of the manufacturing costs of a PM traction motor)
- Small and medium sized applications are usually processed by shredding
- Large products are usually (manually) pre-processed
- Full automatization of dismantling is currently not common

The spectrum of electric machine recycling ranges from manual dismantling to fully automated processing.

Semi-manual processing of medium sized motors

Material input



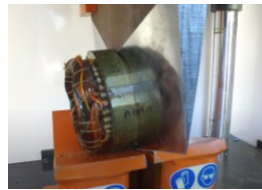
Material output



Typical process flow



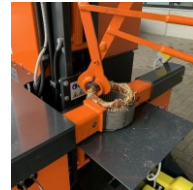
Hydraulic opening of housing



Hydraulic splitting of stator



Hydraulic clamping



Hydraulic removal of windings

Graphics: Bronneberg

- Small scale: semi-manual processing
 - Low output
 - Low investment costs
- Operators: collectors / scrap yards

Fully automatized processing based on shredding

Material input

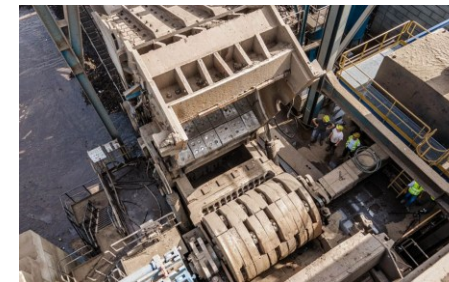


Graphic: Euregio Recycling

Material output



Graphic: Euregio Recycling

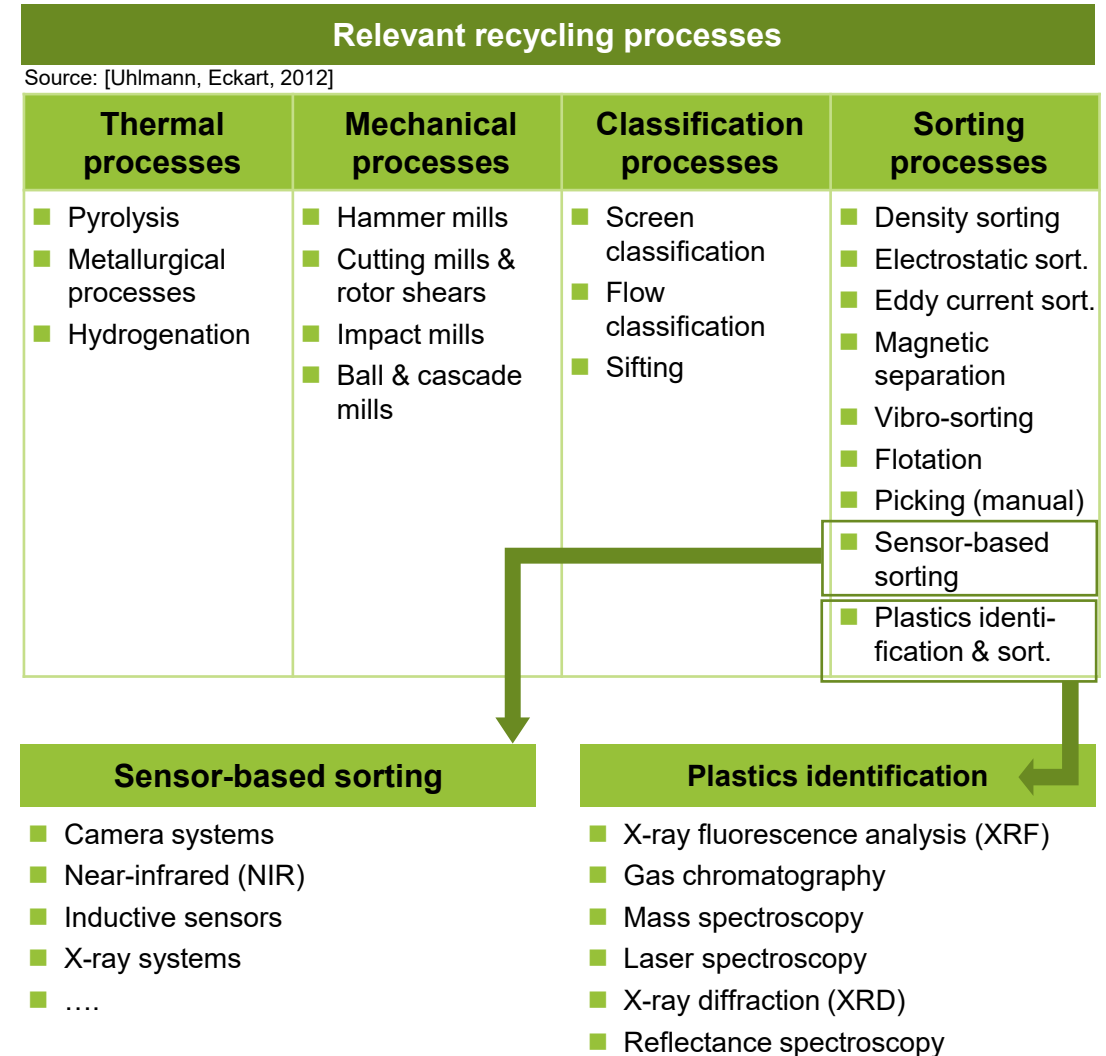
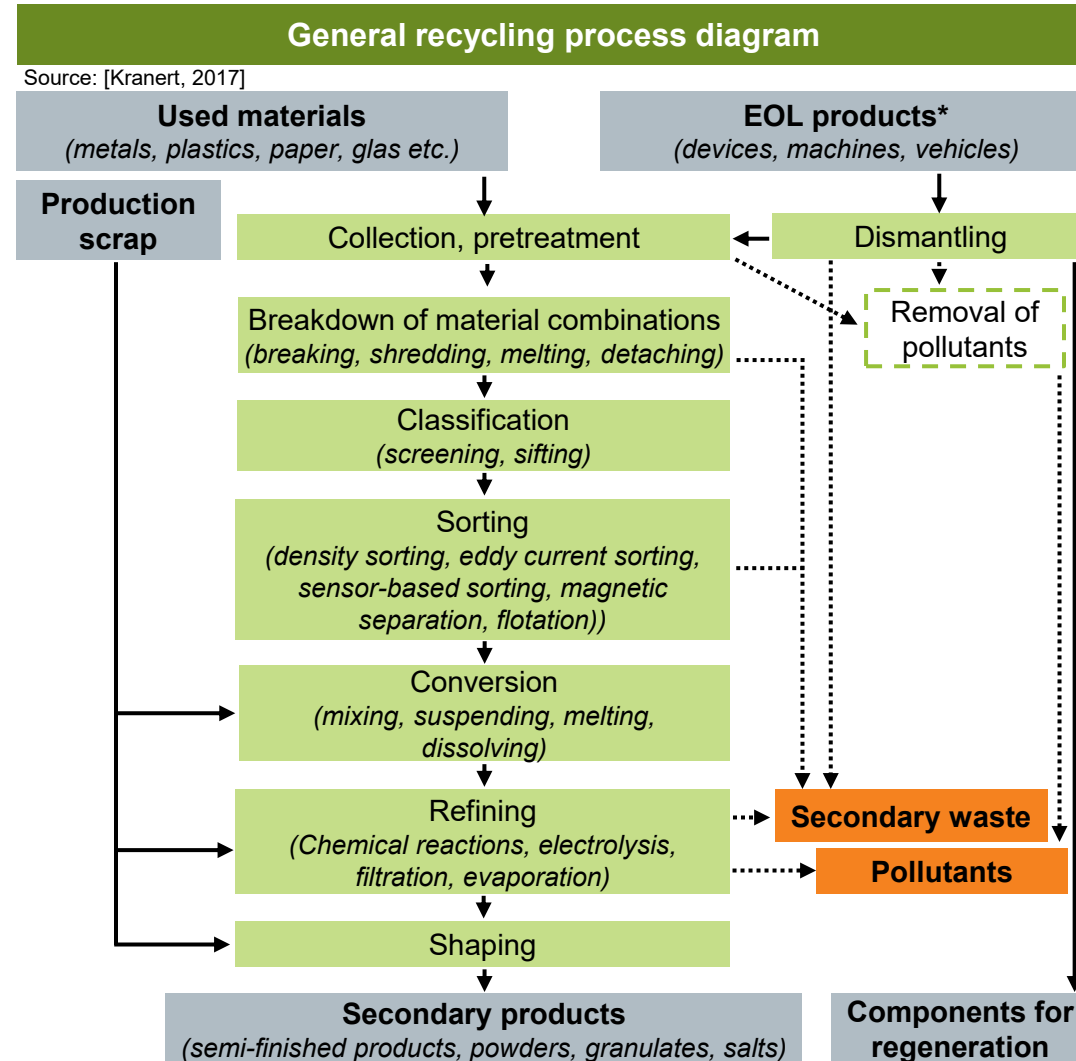


Graphic: Lindemann

Power: 7,500 kW
Capacity: 460t/h

- Industrial scale: automatized and continuous process
 - Very high output
 - Very high investment costs
- Operators: specialized metal recycling companies

Recycling is based on a general process chain whose individual process steps are product-specific and highly interdependent.



Shredding is used to destroy material compounds and to produce optimum piece sizes for sorting or direct use.

Break-down

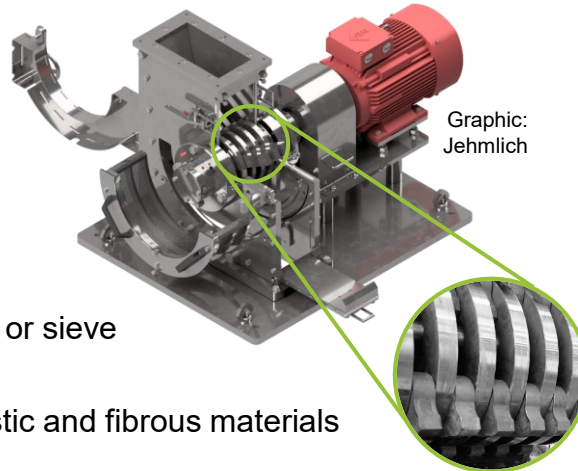
Classification

Sorting

Conversion / Re-fining

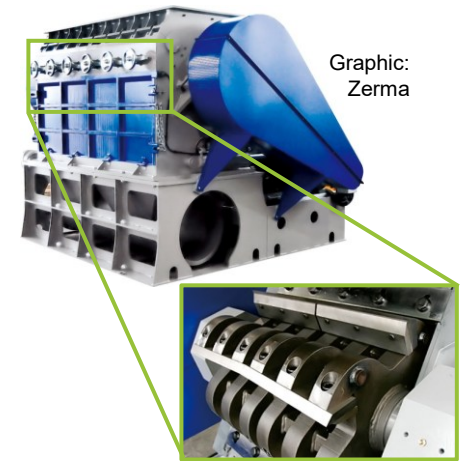
Hammer mill

- **Characteristics:**
High-speed rotors with movable hammers
- **Mode of operation:** Impact crushing
 - Smashing by hammers
 - Crushing on chamber wall
 - Material exiting through grate or sieve
- **Field of application:**
Soft, medium-hard, ductile, elastic and fibrous materials



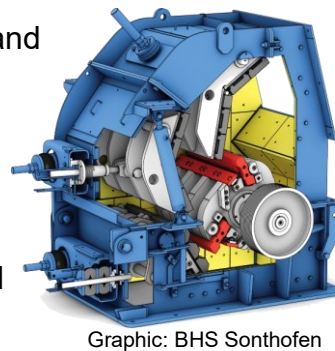
Cutting mill / rotor shears

- **Characteristics:**
Slowly rotating cutting tools
- **Mode of operation:** Shearing
 - Cutting by blades
 - Material exiting through sieve (often pneumatic suction)
- **Field of application:**
 - Soft, elastic and fibrous materials
 - Sensitive to hard components



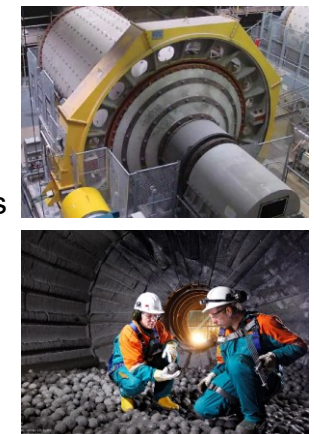
Impact mill / impact crusher

- **Characteristics:**
Rotating roller with interchangeable impact bars and adjustable impact plates in the housing
- **Mode of operation:** Impact crushing
 - Material accelerated by rods
 - Impact on plates and reflection
 - Discharge of material through vibrating channel
- **Field of application:**
Medium-hard and hard materials as well as composite materials



Ball / cascade mill

- **Characteristics:**
Slowly rotating rotary drum with conically inclined front walls & steel balls inside
- **Mode of operation:** Friction and rolling processes
 - Milling by relative movements between steel balls and waste components
 - Homogenization of the material
 - Material leaving through holes in drum
- **Field of application:**
Brittle materials



Shredding is used to destroy material compounds and to produce optimum piece sizes for sorting or direct use.

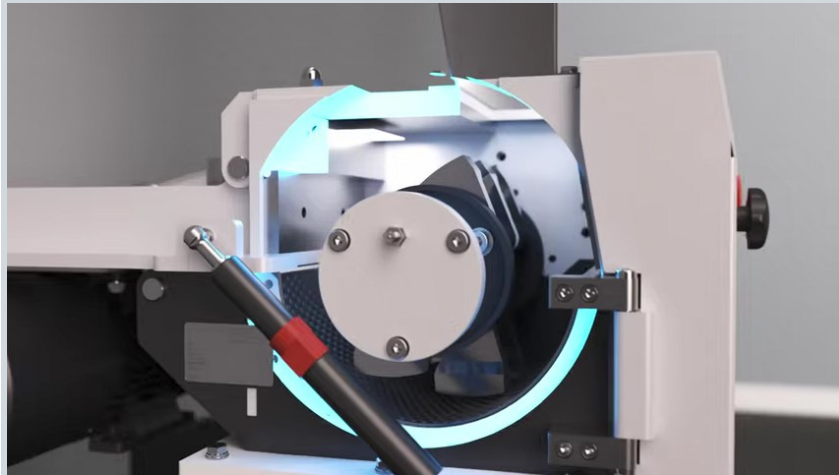
Break-down

Classification

Sorting

Conversion / Re-fining

Hammer mill



Video:
Retsch

<https://www.youtube.com/watch?v=rivrFCuSmHc>

Cutting mill / rotor shears



Video:
Changzhou
Doing Machine

<https://www.youtube.com/watch?v=91cB3gY9KPA>

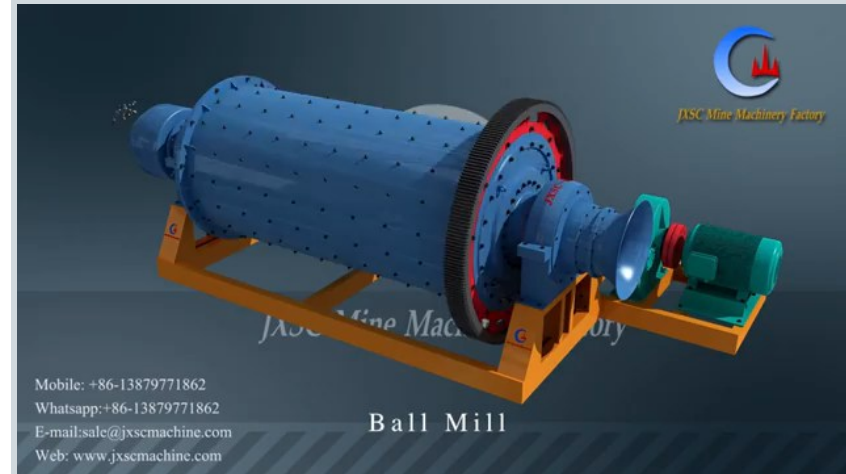
Impact mill / impact crusher



Video:
MESDA Group

<https://www.youtube.com/watch?v=5QaP1pgBGgE>

Ball / cascade mill



Video:
JXSC Mine
Machinery

<https://www.youtube.com/watch?v=qib5O9-kd90>

Classification refers to the separation of a waste mixture into classes of different particle sizes.

Break-down

Classification

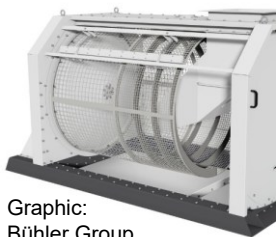
Sorting

Conversion / Re-fining

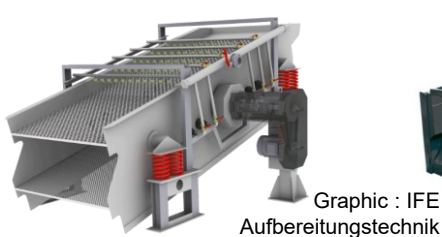
Screening / sieve classification

- **Characteristics:**
Separation based on the characteristic length of the particles by means of a separation surface (screen bottom) with geometrically arranged openings.
- **Mode of operation:**
Classification of grain mixtures according to the particle size
 - Particles passing (sieve passage) the sieve form the fine material
 - Retained particles (sieve overflow) form the coarse material
- **Process variants:**
 - Drum sieve
 - Flip-flop sieve
 - Cup sieve
 - Ballistic separator
 - Disk sieve
 - Linear- & circular vibrating sieves
 - Grizzly sieve
 - Roller sieve
- **Field of application:**
Mixtures of substances with different particles sizes

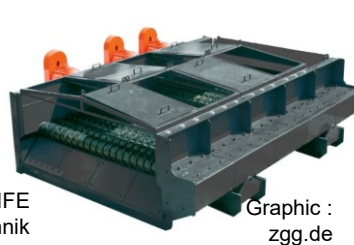
Drum sieve



Flip-flop sieve



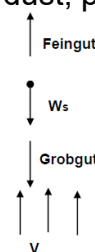
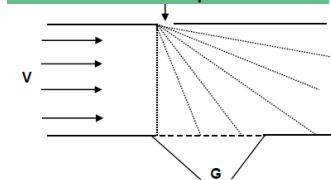
Roller sieve



Flow classification / sifting

- **Characteristics:**
Separation of a particle mixture in a flowing separation medium (air, water, solutions or suspensions)
 - Wind sifting: separation in a gas flow
 - Hydro sifting: separation in a flowing liquid
- **Mode of operation:**
Classification based on the settling velocity of the particles ► influencing parameters:
 - Particle size, geometry & density
 - Working fluid (density, dynamic viscosity)
- **Process variants :**
 - Zigzag windsifter
 - Centrifugal windsifter
 - Horizontal windsifter
 - Cross flow classifiers
- **Field of application:**
Separation of mixtures with significant density differences (e.g. dust, plastics)

Crossflow separation



Counterflow separation



Classification refers to the separation of a waste mixture into classes of different particle sizes.

Break-down

Classification

Sorting

Conversion / Refining

Process

Wind sifting

Video: Pal

<https://www.youtube.com/watch?v=PzLwmUIHKAQ>



Sorting is based on the differing physical properties of the solid particles to be separated.

Break-down

Classification

Sorting

Conversion / Re-fining

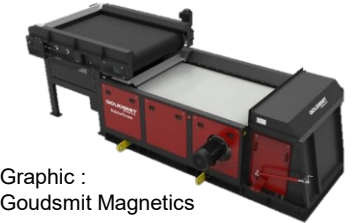
Magnetic separation

- **Characteristics:**
Separation of the ferromagnetic fraction of a material mixture based on the individual magnetizability (magnetic susceptibility)
- **Mode of operation:**
 - Material feed via drums or belts
 - Separation with electromagnets or permanent magnets
- **Process variants:**
Overbelt separators, drum separators, belt separators
- **Field of application:**
Separation of ferrous metals



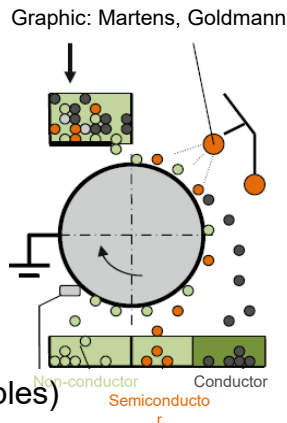
Eddy current separation

- **Characteristics:**
Deflection of conductive particles by electromagnetic induction
- **Mode of operation:**
 - Introduction of particles into a time-changing magnetic field
 - Induction of eddy currents in conductive objects
 - Formation of opposing magnetic fields (Lenz's law)
 - Resulting force causes deflection
- **Field of application:**
Selective separation of non-ferrous metals



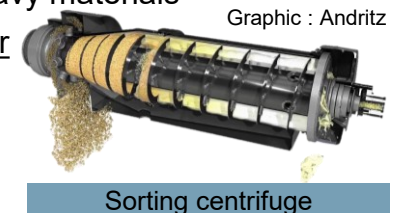
Electro-static separation

- **Characteristics:**
Separation of particles based on the different electric conductivity of different materials
- **Mode of operation:**
 - Surface charging of particles by an electric field or by friction
 - Different deflection due to a superposition of Coulomb force and gravity or centrifugal force
- **Field of application:**
Separation of plastics and metals (e.g. shredded cables)



Density sorting

- **Characteristics:**
Utilization of the density differences between different solids
- **Mode of operation for different methods:**
 - Float-Sink-Sorting: Introduction of material in liquid with def. density
 - Floating of light materials / sinking of heavy materials
 - Hydrocyclone / Sorting Centrifuge: Granular mixture is separated in a centrifugal field
- **Field of application:**
Plastics, Aluminum



Thermal and chemical processes can be used to break down different material composites to obtain pure solids.

Break-down

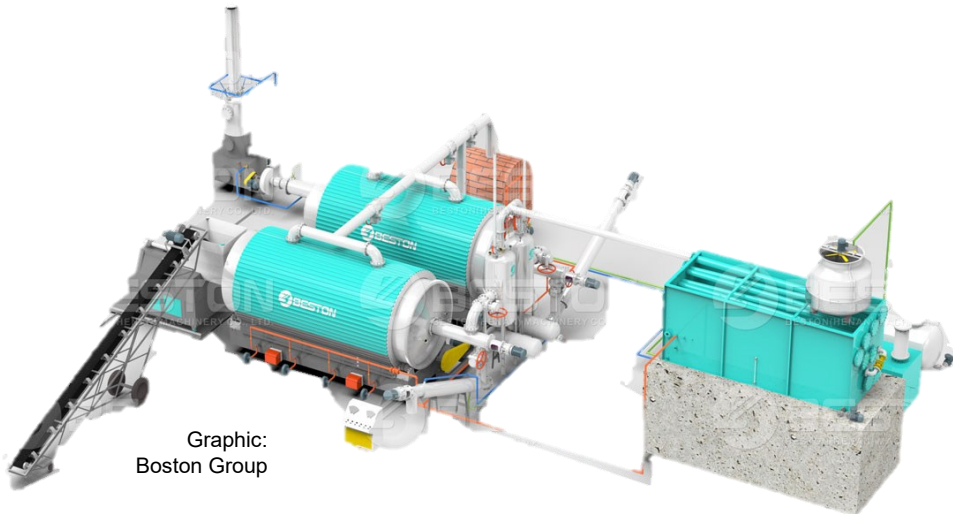
Classi-
fication

Sorting

Con-
version
/ Re-
fineing

Thermal processes: pyrolysis

- **Characteristics:**
Decomposition of solid or liquid substances at high temperatures (400 - 700°C) in the absence of oxygen
- **Mode of operation:**
 - Splitting of organic compounds
 - Embrittlement of composites
 - Melting of e. g. soldered joints
- **Field of application:**
 - Recycling of electronics (e. g. recycling of circuit boards)
 - Recovery of rare earth magnets

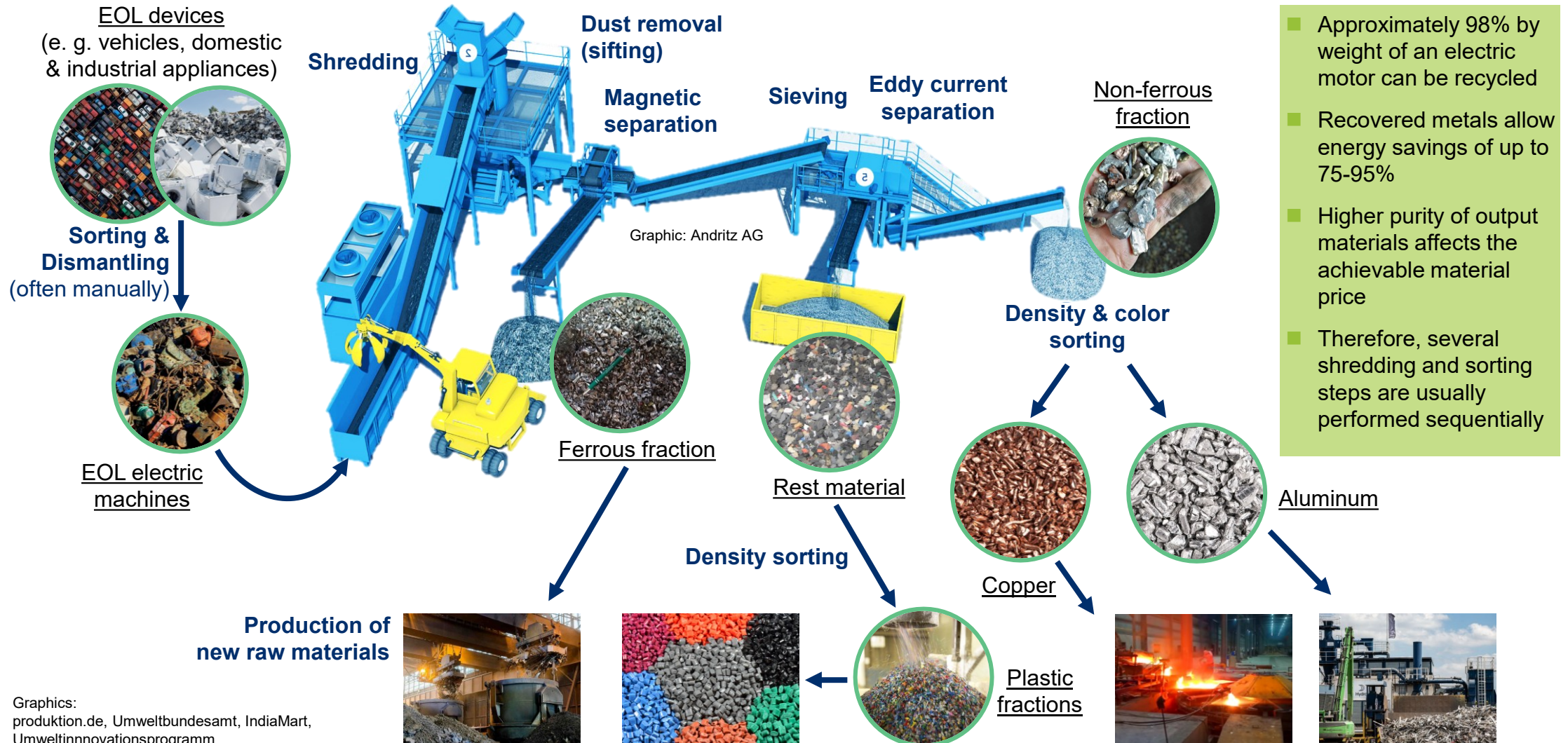


Chemical processes: Extraction electrolysis

- **Characteristics:**
Extraction of metals by leaching metal-containing compound fractions in chemical digestion agents (acids, bases, salt solutions)
- **Mode of operation:**
 - Applying direct current ► Forcing a redox reduction
 - Influences on the leaching process:
 - Physical-chemical properties of the material to be leached
 - Concentration of the solvent
 - Process temperature
 - Leaching time
- **Field of application:**
 - Extraction of precious metals from printed circuit boards
 - Recovery of rare earth magnets



Most electrical machines are recycled with a high degree of automatization by shredding and sorting, whereby the sorting quality has a major influence on the economic efficiency.

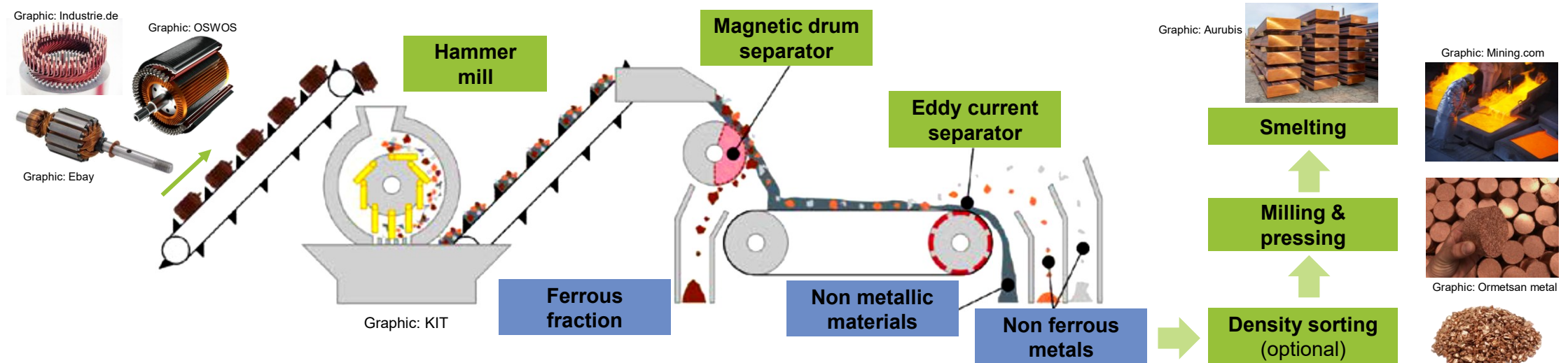


- Approximately 98% by weight of an electric motor can be recycled
- Recovered metals allow energy savings of up to 75-95%
- Higher purity of output materials affects the achievable material price
- Therefore, several shredding and sorting steps are usually performed sequentially

Graphics:
produktion.de, Umweltbundesamt, IndiaMart,
Umweltinnovationsprogramm

Copper is generally the main source of value in EOL traction drives, whereby the degree of purity of the output material has a decisive influence on the economic efficiency.

Typical process chain for copper recycling (low degree of contamination)



Characteristics

- Copper content and location in electrical machines varies by motor type
- Copper can be recycled without loss of quality and recycling saves up to 85% of energy compared to primary production.
- The European recycling rate (share of secondary raw materials in total demand) is approximately 40 - 50%.
- Copper scrap is categorized into quality classes, which influence the price that can be achieved.

Technical challenges

- Contamination (e.g. from coating / insulation system) is common and must be completely removed ► Depending on the degree of contamination, further processing steps may be required before smelting (e.g. pyrolysis or electrolysis)
- Coarse shredding often leads to deformation of the ductile iron cores, making it impossible to separate the copper ("meatballs") ► Several shredding steps are performed sequentially
- Cables require an adapted process chain (high plastic content)



Graphic: FAPS



Graphic: Stokkermill

The market situation for rare earth magnets is endangering the green transition of the European Union.



Situation of demand

- The demand for rare earth (RE) magnets is strongly growing
- Main drivers are „green“ technologies:
 - E-mobility
 - Renewable energies
 - Motors for industrial and domestic applications
- The required rare earth elements (Nd, Tb, Dy) are essential for the industrial strategy of the European Union



Situation of supply

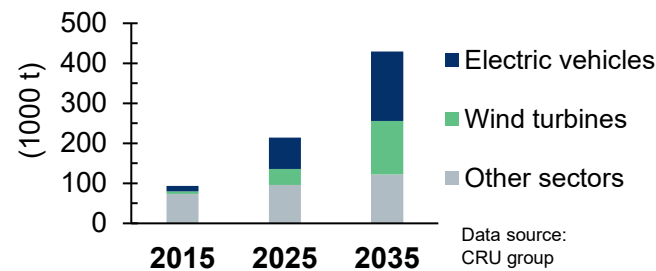


- The PR China is dominating the complete value chain for rare earth magnets
 - Market share for RE-mining: 60 %
 - Market share for RE-refining: 87 %
 - Market share for RE magnet production: 94 %
- China is using market power as political lever
- Primary production in China has a high ecological footprint

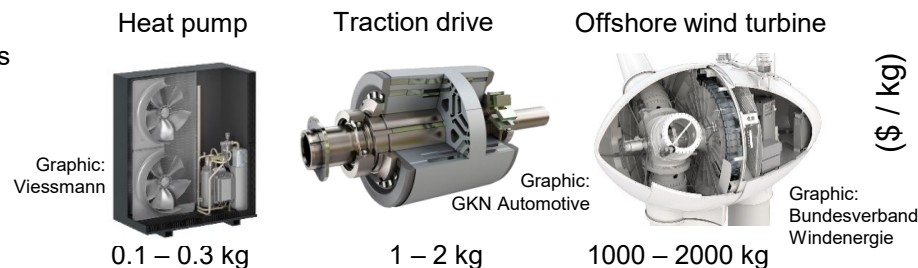
- Potential shortages (demand will exceed supply)
- Cost impact by continuously rising prices
- Poor environmental impact of products

- High supply chain risks due to political interventions
- Price volatility due to speculation

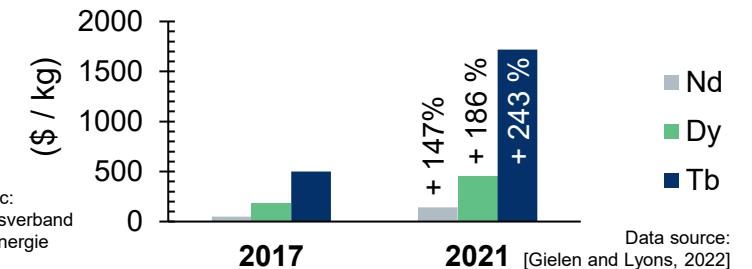
Forecast of global NdFeB magnet demand



RE magnet content in relevant application



Price development of relevant RE oxides



There are currently no industrialized processes available for systematic recycling of rare earth magnets causing a high need for research.

Obstacles and challenges for the recycling of rare earth magnets

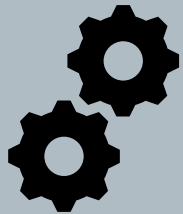
Economic and organizational



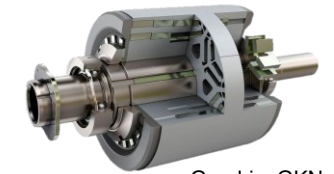
- Economic competition with primary production
- Lack of business cases / difficult to close material cycle (magnet production primarily in China)
- Difficult identification of rare earth content / lack of information (e. g. no labeling)
- Low RE concentration in EOL products / material stream
- No awareness of material value in recycling industry



Technical



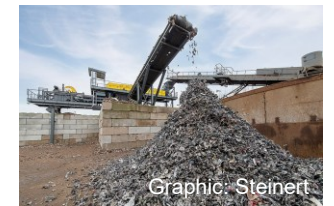
- Properties of EOL products:
 - High variance
 - Lack of dismantlability (installation position, connections) ► high effort for dismantling
 - Condition of EOL products (damages, dirt)
- Properties of rare earth magnets / condition of EOL magnets:
 - Low corrosion resistance, brittleness
 - Magnetic reaction forces (► demagnetization required)
 - Contamination by coatings, adhesives, etc.
- Lack of sufficient recycling technologies:
 - Magnet extraction by dismantling is only economically feasible in rare cases
 - Sorting is difficult (similar properties of hard and soft magnetic materials)
 - currently no solution for shredded material available
 - Currently no industrialization of chemical / physical processes for RE recovery with relevant capacities



Graphic: GKN



Graphic: Recycling magazine



Graphic: Steinert

Less than 1% of RE magnets are currently recycled in the European Union

Various process alternatives are possible for the recycling of rare earth magnets, which have different degrees of technological maturity.

Recycling approaches for rare earth magnets

Development of adapted strategies for specific EOL applications

Extraction strategy

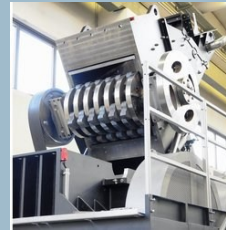
Dismantling



Graphic: Adobe Stock

- Destructive / non-destructive
- Given feasibility
- Cost-effectiveness only for specific applications

Shredding and sorting



Graphic: Bronneberg

- Sorting currently not yet reliably feasible
- Universal applicability

Hydrogen decrepitation



Graphic: Fraunhofer IWKS

- Feasibility proven
- Low throughput

Use of the recovered material

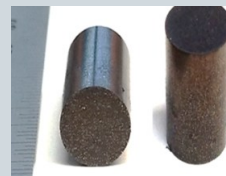
Reuse



Graphic: University of Nottingham

- Use for same application
- Rarely applicable but already practiced

Direct use (short loop)



Graphic: U.S. Department of Energy

- EOL magnets are crushed to powder
- Use for plastic bonded magnets (► downcycling)

Material recycling (long loop)



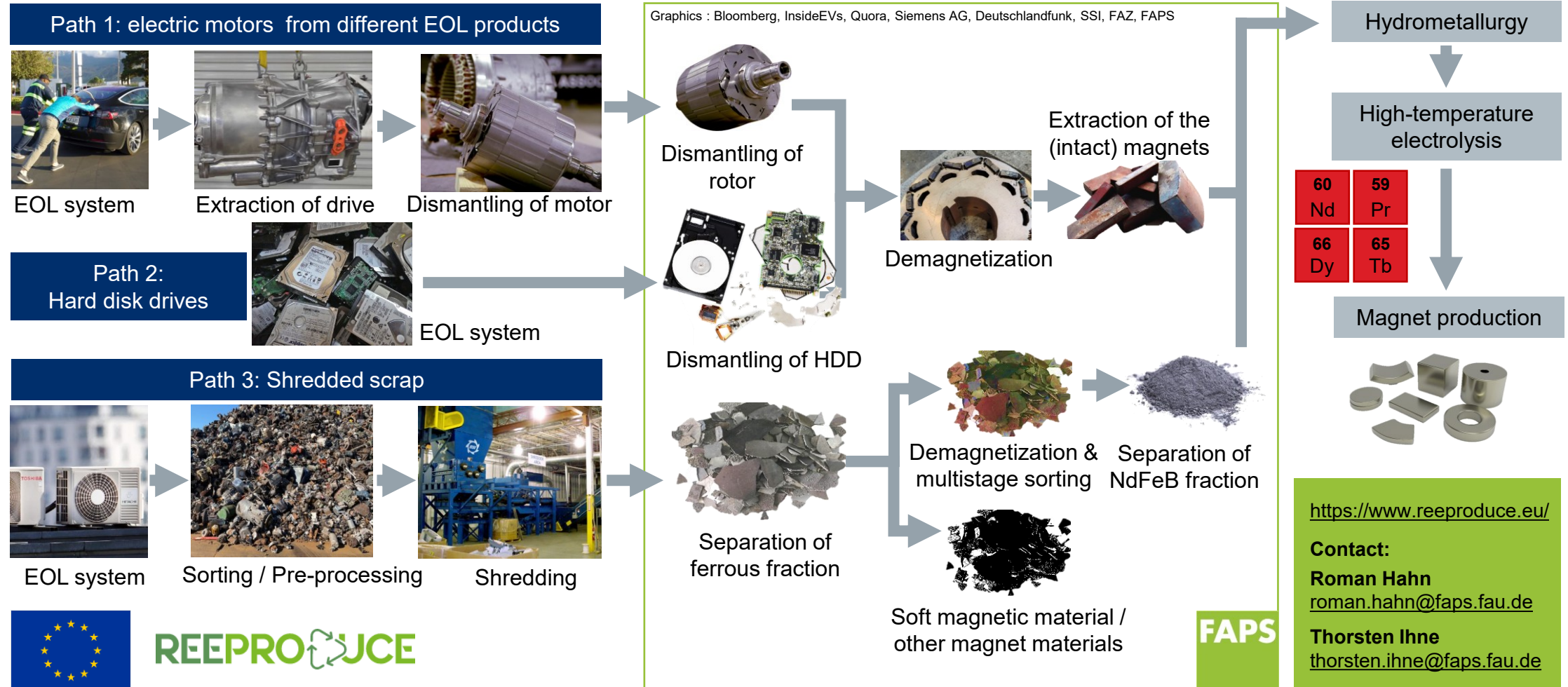
Graphic: Stanford Magnets

- Recovery of REE by pyro- or hydro-metallurgy
- Production of new alloys
- No limitation regarding quality

In parallel the creation of the **organizational framework** conditions is essential (e. g. labeling obligation, mandatory recycling shares, etc.)

As part of the EU funded project REEPRODUCE, FAPS is developing dismantling and sorting processes for the recovery of NdFeB magnets from different sources.

REEPRODUCE is covering the complete process chain from an EOL system to a new magnet.



This lecture unit provides an overview of the motivation and technologies for processing of end-of-life electric machines.

- General overview of circular economy and definitions
- Recycling of electric machines
- **Reuse, repair and refurbishment of electrical machines**
- Design for Recycling



Graphic: Jansen Recycling

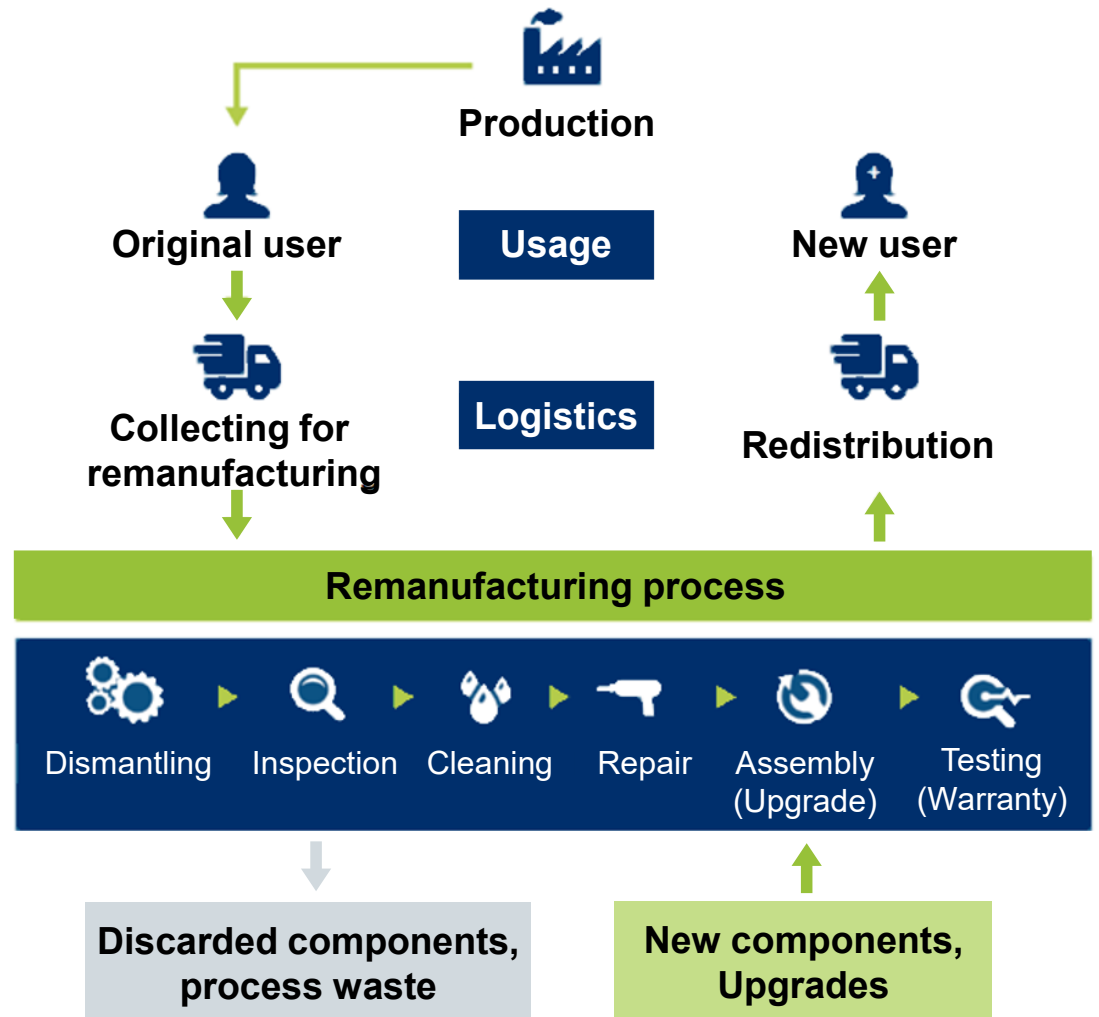
Reuse and remanufacturing allow a significantly extended product life while maintaining the product quality.

Definition

“Remanufacturing involves dismantling the product. By recombining new and reconditioned components and parts, products that are as good as new are manufactured.”
(VDI Guideline 2343)

Characteristics of remanufacturing:

- Is a core element of the circular economy
- Allows products to be reused
- Restores a used device to at least its original condition
- Extends the warranty of the product
- Product is as good as or better than a new product
- From the customer's point of view, the product can be considered equivalent to a new product (► Difference to refurbishment)



The establishment of reuse and remanufacturing is primary driven by pricing policy, environmental protection and customer demand.

Motivations for repair, reuse and remanufacturing within the European Union



Price-Policy

- Rising demand for raw materials and energy
- Rising component and raw material prices due to increasing demand



Environmental protection & climate change

- The consequences of climate change can affect supply chains (e. g. rising sea levels, extreme weather events)
- Stricter environmental regulations
- Higher standards for new products



Customer demand

- Customers demand more environmentally friendly products
- Customers demand lower prices

Remanufacturing in the European Union is hampered by several obstacles, which can be divided into four categories.

Main obstacles for remanufacturing in the circular economy



Product design

- Technical details not always available
- Necessary knowledge for re-X approaches is not made available
- Product design is not adapted to remanufacturing



Competition by cheap products

- The sale of cheap and inferior products reduces the market
- Affects almost all sectors of the circular economy



Additional costs

- High transportation costs for large and bulky products
- High warehousing costs to store products prior to remanufacturing

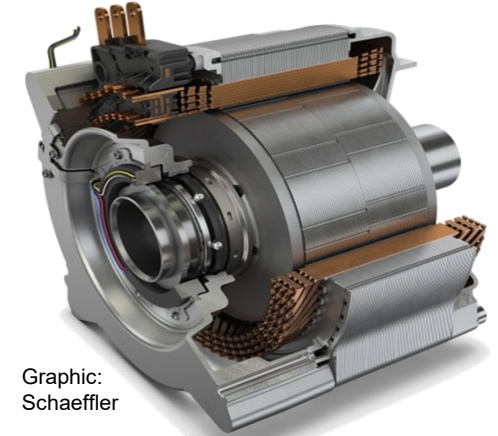


Legal uncertainties

- No clear boundaries yet for the reuse of parts
- Not yet regulated whether products must be labeled as "second-hand"

Non-destructive disassembly is a core competence for establishing remanufacturing.

- The right Re-X approach must be selected based on the specific product condition
- Remanufacturing is currently used primarily in low-volume applications where manufacturing is predominant (e.g. large motors and generators)
- For high-volume applications (e.g. automotive traction drives), automated processes need to be developed (especially for non-destructive disassembly)
- Artificial intelligence has great application potential for Re-X approaches



Graphic:
Schaeffler



Graphic: Borg

Standard process chain

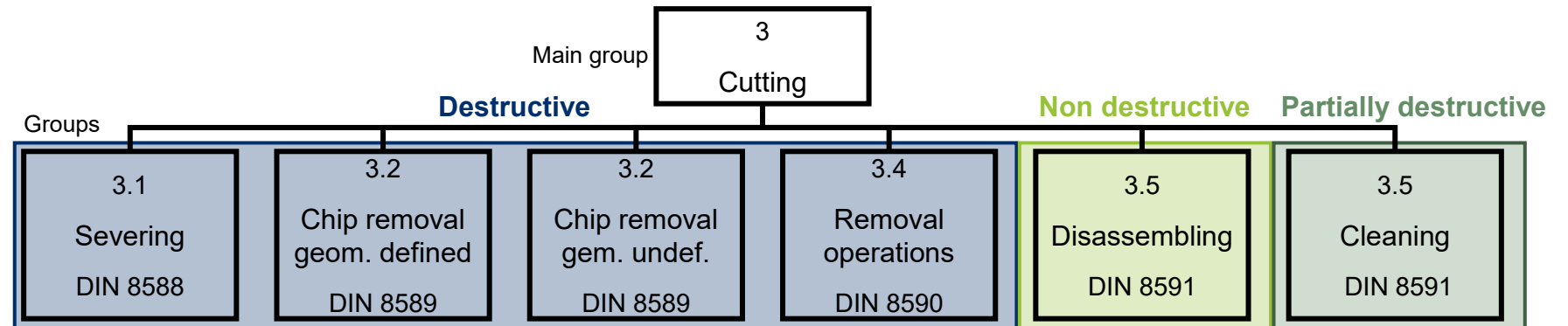
1. Receiving inspection
2. Disassembly
3. Demagnetization (in case of PM motor)
4. Cleaning of components
5. Diagnostics of the components
6. Reprocessing / replacing of components
7. Assembly
8. Testing

Due to various factors, disassembly is usually not the logical inversion of assembly and is difficult to automate.

Definition of disassembly:

"The total of all (intended) processes serving to single out sub-assemblies, components and / or unshaped matter from multi-component systems by separation." (VDI Guideline 2343)

Classification of manufacturing processes according to DIN 8580:



Objectives of disassembly

- Recovery of materials, assemblies and components
- Isolation of valuable and harmful substances
- Conservation of natural resources
- Reduction of landfill demand

Source: [Uhlmann, Eckart, 2012]

Challenges for disassembly ► High flexibility of the disassembly system used required

- Products are not designed for disassembly
- Sometimes very small batch sizes
- Unknown product condition
- Damage
- Influences of soiling, corrosion, ageing & wear
- Difficult to obtain data
- Handling of flexible components
- Complex movements
- Detachable and non-detachable joints in one system
- Non-destructive extraction of usable components
- Fast separation of recyclable materials
- Removal of environmentally hazardous substances

Source: [Uhlmann, Eckart, 2012]

Completely automated disassembly of electric machines is generally difficult to realize and only economically feasible in exceptional cases.

Disassembling is either a preparatory step in recycling or the basis for remanufacturing and reuse

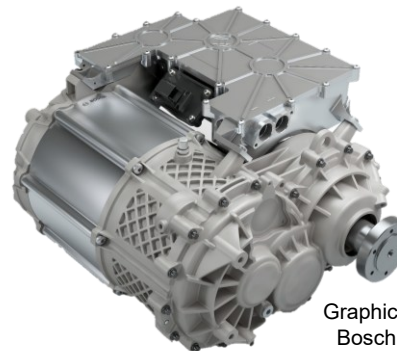
Motivation for full or partial dismantling of electric machines:

- Remanufacturing of overall product
- Reuse of subsystems
- Removal of pollutants or harmful subsystems
- Preparation for recycling:
 - Separation of assemblies for separate processing
 - Simplification of the handling and shredding process
 - Isolation of specific parts that are very difficult to sort out

Application example: automotive traction drives

Factors favoring automated disassembly

- Highly integrated system with several subsystems that are suitable and attractive for remanufacturing
- High product volume
- Limited product variance in the portfolio of a single OEM
- High product and material value
- Intrinsic motivation and development capacities of OEMs
- Established utilization of EOL vehicles
- Existing infrastructure / synergies (e.g. with battery recycling)
- Political regulation of the automotive industry



Graphic:
Bosch

OEM:
Original Equipment Manufacturer

Factors hindering automated disassembly

- Automotive quality requirements hinder reuse
- Complex disassembly leads to low cycle times
- Non-detachable joints / high variance and poor accessibility of joints
- Difficult to handle due to size, weight and geometry
- Magnetic reaction forces on PM motors
- Poor product condition (damage, corrosion, contamination)
- Product variance between OEMs
- Difficult logistics
- Current non-destructive disassembly often not planned

The automated disassembly of automotive traction drives can play a pioneering role in the development of non-destructive disassembly.

Requirements for disassembly of automotive traction units

- High degree of automatization for applicability in high-wage countries
- High flexibility due to product variance and product condition
- Careful handling of components
- Short cycle times



Graphic:
Valeo



Graphic: FAPS

Potential solution for automotive traction unit

- Condition assessment to determine further processing (remanufacturing, recycling or disposal)
- Automated disassembly (level of automation < 100%)
- System with several stations (separation of subsystems, dismantling of subsystems)
- Robot systems in combination with vision systems and tool changing stations

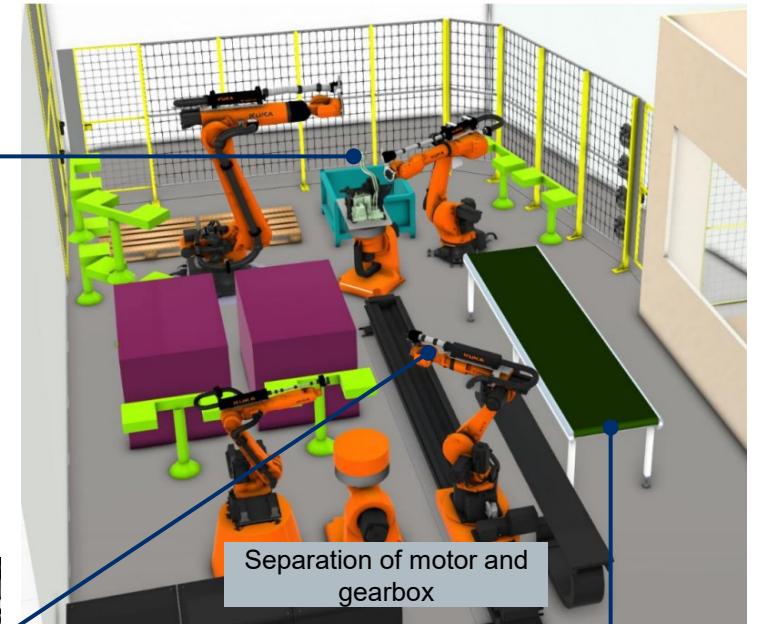
Separation of rotor
and stator



Dismantling
screwdriver



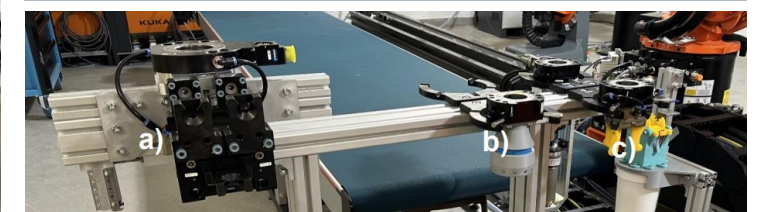
Dismantling cell for automated processing
of EOL traction units



Separation of motor and
gearbox

Graphics: KIT

Tool changing station



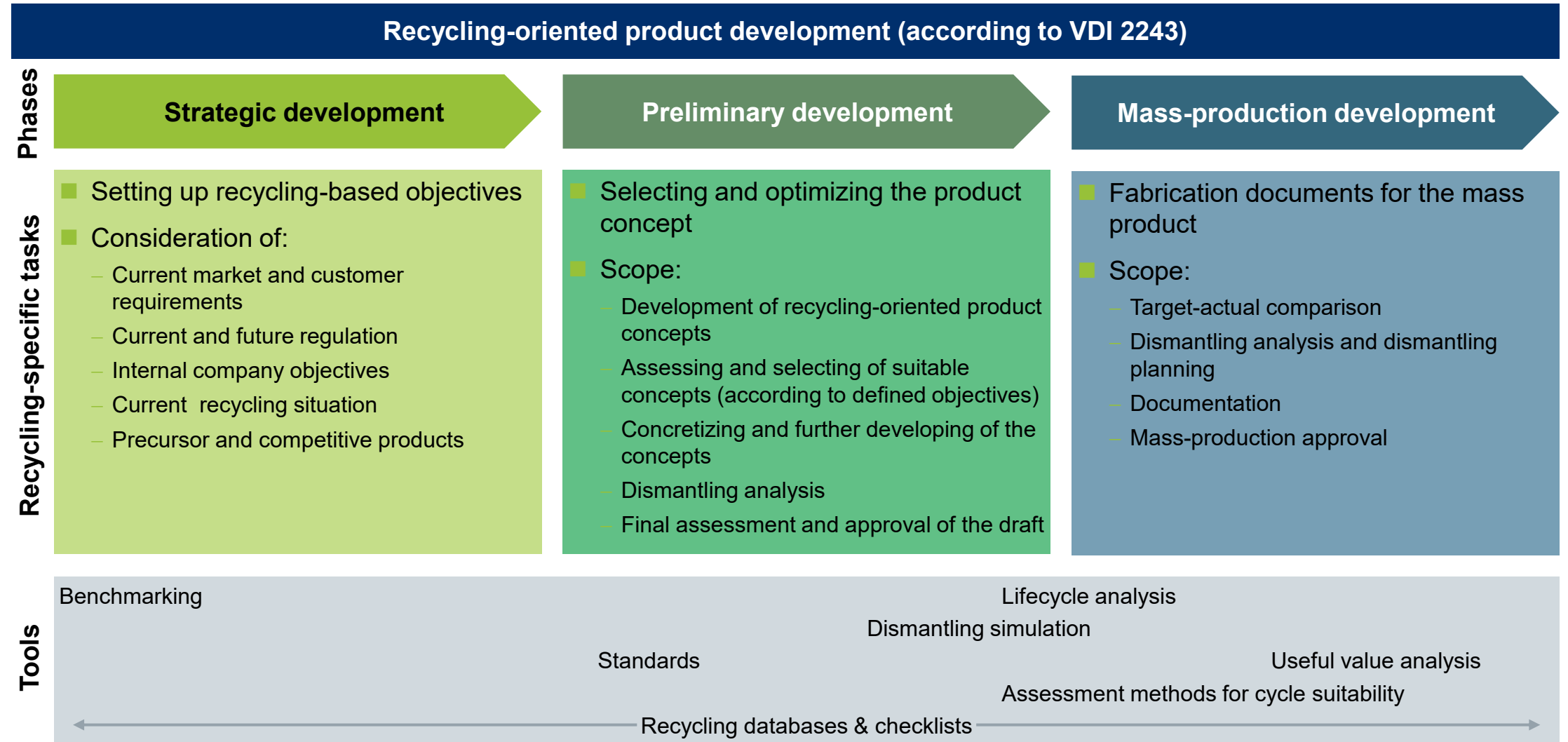
This lecture unit provides an overview of the motivation and technologies for processing of end-of-life electric machines.

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- **Design for Recycling**



Graphic: Jansen Recycling

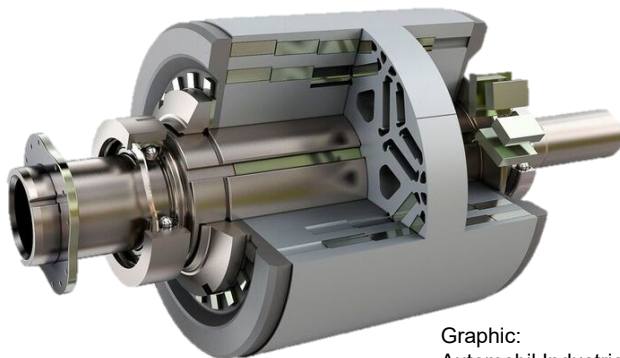
Design for Recycling considers the suitability for Re-X approaches already during the three phases of the product development process.



The design process must ensure that the product is suitable for disassembly and material recovery.

Material compliance

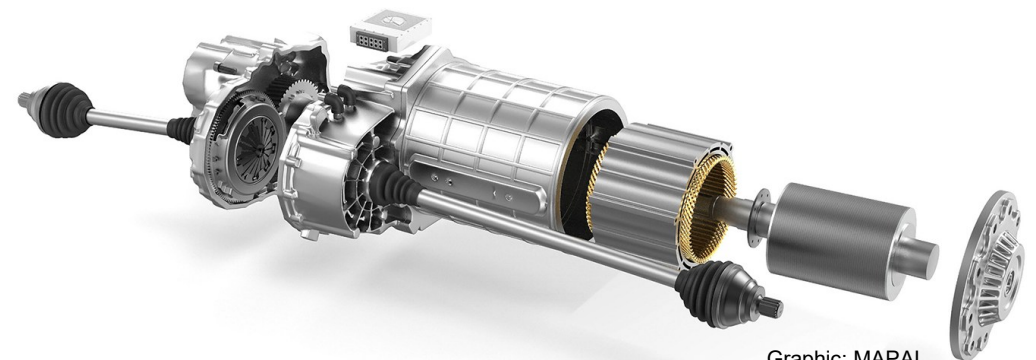
- Goal: minimizing the use of materials
- Objectives
 - Ability for identification (e.g. by clear labelling)
 - Recyclability of materials (multiple reusability)
 - Recycling compatibility in material compounds
 - Avoidance of environmentally critical contents, which have to be removed specifically and eliminated separately
 - Avoidance of recycling-critical materials, which disrupt specific recycling processes (e.g. transmission oil in the shredder process)
 - Simple disposal of wastes



Graphic:
Automobil Industrie

Disassembly compliance

- Goal: simplicity of disassembly
- Objectives:
 - Adapted product structure
 - Visual perceptibility of relevant components (e.g. batteries)
 - Accessibility (e.g. direct access)
 - Detachability (e.g. non-destructive / destructive)
 - Variance and detachability of connections
 - Disassembly and breakdown time



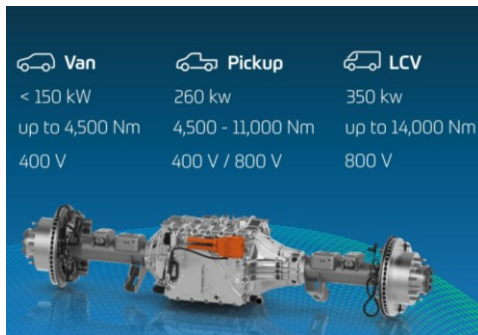
Graphic: MAPAL

Product design guidelines can make it easier to remanufacture and reuse products.

Design guidelines

Modular design

- Increased flexibility
- Increased scalability
- Simplifies the assembly process
- Minimizes manufacturing costs



Graphic: ZF

Retrofit ability

- Hardware is adaptable
- Future-proofing
- Increased sustainability



Graphic : ZF

Standardization

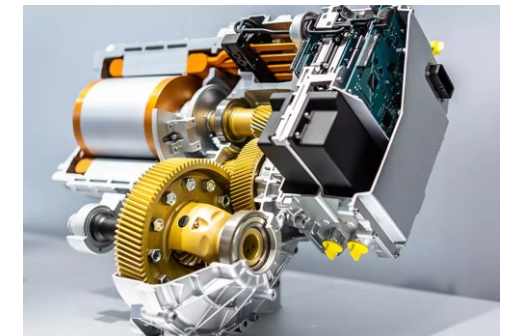
- Enable reusability
- Reduce waste
- Future-proofing
- Improved interoperability



Graphic: Phoenix Contact

Use of high-quality materials

- Increased reliability
- Improved safety
- Enable the reusability
- Economic incentive to recycle



Graphic : Viona Magnetics

In addition to the choice of material, the design of the joints plays a key role for the design for recycling approach.

Classification of joining processes according to DIN 8593

Good solvability

Only destructively solvable

Putting together

Filling

Pressing on / in

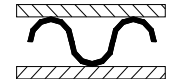
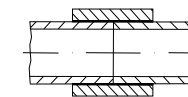
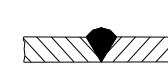
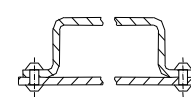
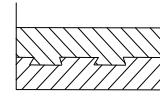
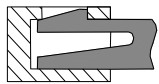
Shapeless forming

Forming

Welding

Soldering

Bonding



Screwing

Clamping

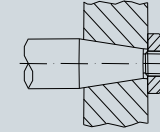
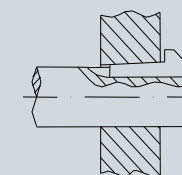
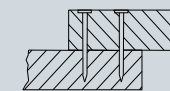
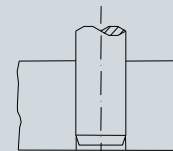
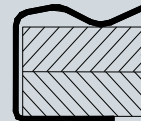
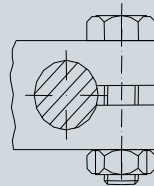
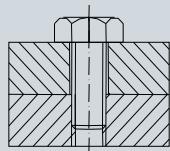
Stapling

Pressing

Nailing / hammering

Wedging

Bracing



Joining according DIN 8593-0: „The permanent joining or other bringing together of two or more workpieces of a specific geometric shape or of similar workpieces with shapeless material “

Source: DIN 8593-0:2003-09

The contents of the lecture unit can be further deepened with the following specialized literature:

Books

- Martens, Hans; Goldmann, Daniel (2016): *Recyclingtechnik*. Wiesbaden: Springer Fachmedien Wiesbaden.
- Kranert, Martin (Hg.) (2017): *Einführung in die Kreislaufwirtschaft. Planung - Recht - Verfahren*. Springer Fachmedien Wiesbaden. 5. Auflage. Wiesbaden, Heidelberg: Springer Vieweg.

Other publications

- VDI 2243. *Recycling-oriented product development*
- Bast, Ulrich; Blank, Rolf; et al. (2014): *Recycling von Komponenten und strategischen Metallen aus elektrischen Fahrantrieben. Abschlussbericht zum Verbundvorhaben. Kennwort: MORE (Motor Recycling)*. Available at: <https://edocs.tib.eu/files/e01fb15/826920594.pdf>
- Baazouzi Sabri; Rosenberg, Sonja; et al. (2023): *Industrielle Demontage von Batteriemodulen und E-Motoren zur Sicherung wirtschaftsstrategischer Rohstoffe für die E-Mobilität – DeMoBat. Abschlussbericht*. Available at: <https://www.ipa.fraunhofer.de/content/dam/ipa/de/documents/Projekte/Industrieprojekte/DeMoBat-Abschlussbericht.pdf>, zuletzt geprüft am 12.01.2024.

Essays

- Yang, Yongxiang; Walton, Allan; et al. (2017): REE Recovery from End-of-Life NdFeB Permanent Magnet Scrap: A Critical Review. In: J. Sustain. Metall. 3 (1), S. 122–149. DOI: 10.1007/s40831-016-0090-4.





**Was hat Ihnen gefehlt, um die
Inhalte noch besser zu
verstehen?**



FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

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**Friedrich-Alexander-Universität
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THANK YOU